Software Engineering Design Capstone Project Proposal

Offline-First Disaster Relief Distribution & Coordination System

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Semester: Fall 2026

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Course Code: SE331

Course Name: Software Engineering Design Capstone Project

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CHAPTER 1: INTRODUCTION

In August 2024, flash floods in the northeastern and southeastern regions of Bangladesh affected about 4.94 million people. Many volunteer groups collected relief and carried it into the flooded districts, often working separately with no shared view of where others had already been. Residents reported that households near roads received relief several times while families in remote areas received little or nothing. Poor connectivity made the problem harder to solve. On 24 August, 1,807 cell towers were out of service, and communication in Feni was reported as fully disrupted. A tool that needs a live internet connection is least reliable in the places where relief is actually handed out.

Existing relief management tools are designed around a single organisation's own programme, so they do not give many small, independent groups a shared household-level record to work from. The Offline-First Disaster Relief Distribution & Coordination System is proposed to fill this gap. Volunteers use a mobile application to register consenting households and record each distribution without an internet connection. The records are stored on the phone and synchronised with a central server when a connection returns. Coordinators can then see which areas have registered households with no recorded relief.

The main strength of the system is that its offline-first design works together with a household record shared across organisations. Because every participating group writes the same record, the server can compare households registered by different groups and flag possible duplicates for a coordinator to review. Once field records are synchronised, a group arriving in an area can see where relief has already reached and where it has not. The system supports these human decisions and does not decide who receives aid.

1.1 Objective

The main objective of this project is to design and develop an offline-first software system that gives independent volunteer groups one shared, household-level record of relief distribution during a disaster. Field volunteers should be able to register households and record distributions on a mobile device without an internet connection. These records are synchronised with a central server when a connection returns, and each record is stored exactly once even if an upload is interrupted. The system should flag conflicting edits and possible duplicate households across organisations for review by an authorised coordinator and show the relief coverage of each area based on synchronised records. Every action should be kept in an auditable history, and the system will be evaluated through automated tests using simulated devices and synthetic household data.

1.2 Key Features

- **User Accounts and Role-Based Access:** Users register with a mobile number and password and join an organisation. Their account becomes active only after a coordinator of that organisation approves it. Each role can perform only the actions allowed to it, and every request is checked on the server.
- **Operation Setup:** An operation administrator creates a disaster operation and defines its affected areas as upazilas, unions and wards. The administrator also maintains a list of relief items and approves the organisations that join the operation.
- **Offline Household Registration:** Volunteers register households on an Android phone without an internet connection, after reading out a short consent statement in Bangla. Each household receives a short code that later teams can use to find it. Before saving, the app warns the volunteer if a similar household already exists in the data stored on the phone.

- **Offline Distribution Logging:** Volunteers record the items and quantities given to each household. If a family does not consent to registration, relief can still be recorded for the area without personal details. A mistaken record can be cancelled with a reason, but it is kept and marked as voided so the history remains visible.
- **Data Synchronisation:** When the phone has a connection, it uploads waiting records in batches and downloads new records from other groups. An interrupted upload resume from where it stopped, and a record that arrives twice is stored only once. The app always shows how many records are waiting and when the last sync took place.
- **Conflict Detection and Review:** If two volunteers change different details of the same household while offline, both changes are kept automatically. If they enter different values for the same detail, the system marks it for review, and a coordinator chooses the correct value.
- **Cross-Organisation Duplicate Detection:** After each sync, the server compares newly registered households with existing ones in the same union and gives each pair a similarity score based on phone number, name, location and family size. Likely and possible duplicates are sent to a coordinator, who decides whether they are the same household. The system never merges records on its own, and a link between records can be undone.
- **Area Coverage View:** For each area, the dashboard shows how many registered households have received relief, how many have not, and when the last distribution was recorded. An area with no registered households is labelled as having no records, so that it is not mistaken for an area where nobody needs help.
- **Notifications:** Coordinators are notified when a new duplicate or conflict involving their organisation is found. Volunteers are notified when records have been waiting too long to sync or when a sync fails.

- **Audit Trail and Reporting:** Every registration, distribution, correction and review decision is recorded with the user, organisation, device and time, and cannot be edited or deleted through the system. Coordinators can produce reports of their organisation's distributions for donors, and operation-wide summaries are available without any names or phone numbers.
- **Privacy and Data Security:** The system collects only the household details needed for coordination and never stores national ID numbers or photos of people. Data on each phone is encrypted, phone numbers are masked for users outside the registering organisation, and personal details are deleted 90 days after an operation closes.

CHAPTER 2: SYSTEM ANALYSIS

2.1 Scenario Writing

2.1.1 Scenario 1: Register a Household Offline

Scenario Description:

- While still online, download the household data for the target union to the field app
- Travel to the affected area, where there is no mobile network
- Read the consent statement aloud in Bangla and record whether the household agrees
- Enter the head of household's name, ward and family size
- If the phone shows a warning about a similar household, either pick the existing record or continue with the new one
- Save the household on the phone and write its short code on a card for the family

2.1.2 Scenario 2: Record a Relief Distribution

Scenario Description:

- Look up the household on the phone by short code, name or mobile number
- Choose the relief items given and enter the quantity of each
- Note whether the family has already received relief from another group that day
- Save the distribution record on the phone
- If the family declines to register, record an anonymous distribution for the area instead

2.1.3 Scenario 3: Synchronise After an Interrupted Upload

Scenario Description:

- Move to a place with mobile signal and start a sync from the field app
- Upload the waiting records to the server in batches
- The connection drops partway through the upload

- When the signal comes back, start the sync again
- The upload resumes from the first record the server has not confirmed, so no record is stored twice
- Download the new records from other groups and check the updated sync status

2.1.4 Scenario 4: Review a Possible Duplicate Household

Scenario Description:

- Log in to the coordination dashboard once volunteers from two organisations have synced their records
- Open the notification about a new possible duplicate household
- Compare the two records side by side, with the score for each matching signal
- Mark the pair as the same household or as different households
- If the decision turns out to be wrong, undo the link later

2.1.5 Scenario 5: Resolve a Conflicting Household Edit

Scenario Description:

- Open the conflict list after two offline edits to the same household detail have synced
- Compare the two values, including who entered each one and when
- Pick the correct value, or type in a corrected one
- Save the decision so it is added to the household's history

2.1.6 Scenario 6: Check Area Coverage

Scenario Description:

- Log in to the coordination dashboard
- Open the coverage view for the operation

- For each area, see how many registered households have received relief and how many have not, and when the last distribution was recorded
- Pick out areas with no recorded distribution or no registered households
- Before the next visit, download the list of unserved households in an area to the field app

2.2 Stakeholder

The key stakeholders involved in the Offline-First Disaster Relief Distribution & Coordination System are:

1. **Field Volunteer:** Members of participating relief organisations who use the offline field application to register households, record relief distributions and synchronise their data when a connection becomes available.
2. **Group Coordinator:** The leader of a participating organisation's relief work, who approves its volunteers, reviews possible duplicate households and conflicting edits, and monitors area coverage through the web dashboard.
3. **Operation Administrator:** A trusted person or organisation that sets up a disaster operation by defining its affected areas and relief items, approves the organisations that join it, and closes it when the relief work ends.
4. **System Administrator:** Technical personnel, represented by the development team during the project, who maintain the platform, approve organisation accounts, manage user accounts and devices, and handle data backup and retention.
5. **Observer:** Invited read-only users, such as partner NGO staff or local officials, who view area-level coverage figures and summary reports without access to personal household details.

6. **Beneficiary Household:** Families affected by the disaster who do not use the system themselves, but whose details are recorded by volunteers with their consent and who can receive relief whether or not they agree to be registered.

2.3 User Profile

The characteristics below describe the expected users of the system. Age ranges, education levels and user numbers are working assumptions and will be refined through discussions with volunteer groups.

2.3.1 User Profile-01: Field Volunteer

User Class	Notes on Characteristic	Requirement Implied
Type of User	Field volunteer of a participating relief organisation	Verification, Security
Age Range	Approximately 18 to 35	Interface
Frequency of Use	Many times a day during an active operation	Performance, Reliability, Operational
Mandatory / Discretionary	Discretionary; volunteers and organisations join voluntarily	Acceptance, Interface
Computer Experience / OS / Applications	Basic smartphone use on low-cost Android phones; mainly messaging and social media apps	Interface, Portability, Documentation
Education	Mostly secondary school leavers and university students	Interface, Documentation
Goals	Register households and record relief correctly without a network	Reliability, Performance, Acceptance
Language Skills	Bangla; basic English for entering names in English letters	Interface, Documentation
Number of Users	Tens to hundreds per operation	Performance, Reliability, Resource
Training	A briefing of about 10 minutes before field work	Documentation, Acceptance
Other Systems Used	Facebook groups, messaging apps, phone calls and paper lists	Acceptance
Ways of Working	Outdoors in flooded areas, often offline and one-handed	Reliability, Interface, Security, Safety

2.3.2 User Profile-02: Group Coordinator

User Class	Notes on Characteristic	Requirement Implied
Type of User	Leader of one participating organisation's relief work	Verification, Security
Age Range	Approximately 22 to 45	Interface
Frequency of Use	Several times a day during an active operation	Performance, Operational
Mandatory / Discretionary	Mandatory; each participating organisation needs at least one	Operational, Security
Computer Experience / OS / Applications	Comfortable with smartphones, laptops, web browsers and spreadsheets	Interface, Portability
Education	Usually HSC or above, often university students or graduates	Documentation
Goals	Resolve duplicates and conflicts and find areas where relief has not reached	Reliability, Performance, Acceptance
Language Skills	Bangla and English	Interface, Documentation
Number of Users	One to three per organisation	Security, Operational
Training	Hands-on session on review queues and the coverage view	Documentation
Other Systems Used	Spreadsheets, Facebook groups and messaging apps	Operational, Acceptance
Ways of Working	Web dashboard with internet access; field app when in the field	Portability, Performance, Security

2.3.3 User Profile-03: Operation Administrator

User Class	Notes on Characteristic	Requirement Implied
Type of User	Trusted person or organisation managing one disaster operation	Verification, Security
Age Range	Approximately 25 to 50	Interface
Frequency of Use	Intensive at setup and closing; daily monitoring in between	Performance, Operational
Mandatory / Discretionary	Mandatory; every operation requires one	Operational, Reliability
Computer Experience / OS / Applications	Experienced with web applications, spreadsheets and CSV files	Interface, Documentation
Education	Graduate	Documentation
Goals	Set up the operation correctly, monitor activity and close it when relief ends	Security, Reliability, Operational
Language Skills	Bangla and English	Interface, Documentation
Number of Users	One, or a small team, per operation	Security
Training	Session on operation setup, audit search and reports	Documentation
Other Systems Used	Spreadsheets, email and messaging apps	Operational
Ways of Working	Laptop with a stable connection, usually from an office	Performance, Security

2.3.4 User Profile-04: System Administrator

User Class	Notes on Characteristic	Requirement Implied
Type of User	Technical maintainer of the platform; the development team during the project	Verification, Security
Age Range	Approximately 22 to 40	Interface
Frequency of Use	Daily during active operations; as needed at other times	Reliability, Operational
Mandatory / Discretionary	Mandatory	Operational, Maintenance
Computer Experience / OS / Applications	Advanced; server administration, databases and monitoring tools	Maintenance, Documentation
Education	Degree in Software Engineering or a related field	Documentation
Goals	Keep the platform secure, available and backed up	Security, Reliability, Safety, Maintenance
Language Skills	Technical English and Bangla	Documentation
Number of Users	One or two	Security
Training	Technical documentation on deployment, backup and data retention	Documentation
Other Systems Used	Server, database and monitoring tools	Maintenance, Portability
Ways of Working	Administration section of the dashboard and remote server access	Security, Reliability, Performance

2.3.5 User Profile-05: Observer

User Class	Notes on Characteristic	Requirement Implied
Type of User	Invited read-only user, such as partner NGO staff or a local official	Verification, Security
Age Range	Approximately 25 to 55	Interface
Frequency of Use	A few times a week during an operation	Performance
Mandatory / Discretionary	Discretionary; access by invitation only	Acceptance, Security
Computer Experience / OS / Applications	Basic to moderate; web browser and PDF reports	Interface, Portability
Education	Graduate	Documentation
Goals	Follow coverage by area and obtain summary reports	Security, Acceptance
Language Skills	Bangla and English	Interface
Number of Users	A few per operation	Security
Training	Brief orientation	Documentation
Other Systems Used	Own organisation's reporting tools and spreadsheets	Operational
Ways of Working	Web browser, view only	Security, Reliability

2.4 Scope

1. **User Registration and Access Control:**

   (a) **Objective:** Let members of participating organisations register and use the system according to their role.

   (b) **Functionality:** Users register with a name, mobile number and password, and choose the organisation they belong to. An account becomes active only after a coordinator of that organisation approves it. Each role is limited to the actions it is permitted, and the server checks every request. After an online login, the field application stays usable offline for a limited period, protected by an app PIN.

2. **Operation and Area Setup:**

   (a) **Objective:** Let an operation administrator prepare the system for a specific disaster response.

   (b) **Functionality:** The administrator creates an operation and defines its affected areas as upazilas, unions and wards, either one at a time or from a CSV file. The administrator also keeps the list of relief items, approves organisations that ask to join, and closes the operation when relief work ends.

3. **Offline Field Work:**

   (a) **Objective:** Let volunteers register households and record relief distributions without an internet connection.

   (b) **Functionality:** Before going to the field, volunteers download the data for their selected areas. They register households that consent, get a short code for each one, and can search households by code, name or mobile number. If a similar household already exists on the phone, the application warns the volunteer before saving. Distributions are recorded against registered households, or anonymously for the area when a family declines to register. All records are kept in an encrypted database on the phone.

4. **Data Synchronisation:**

   (a) **Objective:** Share records between devices and the central server whenever a connection is available.

   (b) **Functionality:** The application uploads waiting records in batches, automatically when connected or manually on request. If an upload is interrupted, it resumes from the first unconfirmed record, and a record received more than once is stored only once. During each sync, the phone also downloads new records from other organisations for its selected areas.

5. **Conflict Handling:**
   (a) **Objective:** Stop offline edits from being silently overwritten.
   (b) **Functionality:** Changes to different details of the same household are merged automatically. When two devices enter different values for the same detail, the server records a conflict, and a coordinator chooses or enters the correct value.
6. **Duplicate Detection:**
   (a) **Objective:** Find households that different volunteers or organisations may have registered more than once.
   (b) **Functionality:** After each sync, the server compares new or changed households with existing ones in the same union and calculates a similarity score from the mobile number, name, location and family size. Likely and possible duplicates go into a review queue, where a coordinator decides whether each pair is the same household. Records are never merged automatically, and a link can be undone. The system also flags a household that received the same item from two organisations within 24 hours.
7. **Coverage Analysis:**
   (a) **Objective:** Show coordinators where relief has been recorded and where it has not.
   (b) **Functionality:** For each area, the dashboard shows the number of registered households, how many have received relief, and the date of the last recorded distribution. Areas with no registered households are labelled separately so they are not mistaken for covered areas. Before a visit, coordinators can download the list of unserved households in an area.
8. **Notifications:**
   (a) **Objective:** Tell users about events that need their attention.

(b) **Functionality:** Coordinators are notified of new duplicates and conflicts involving their organisation. Volunteers are notified when a sync fails or when records have been waiting too long to sync.

9. **Audit Trail and Reporting:**

(a) **Objective:** Keep a traceable history of all actions and support reporting to donors and partners.

(b) **Functionality:** Every registration, distribution, correction, review decision and login is recorded with the user, organisation, device and time. These records cannot be edited or deleted through the system. Coordinators can generate reports of their organisation's distributions, and operation-wide summary reports contain no names or phone numbers. Reports can be downloaded as PDF and CSV.

10. **Security and Privacy Measures:**

(a) **Objective:** Protect household data and the integrity of the system.

(b) **Functionality:** All communication uses HTTPS, passwords are stored as salted hashes, and data on each phone is encrypted. The system collects only the household details needed for coordination and masks phone numbers for users outside the registering organisation. Personal details are deleted 90 days after an operation closes.

2.4.1 Scope Boundaries

Area claiming between organisations, need-based priority lists of unserved households and a map view of coverage belong to the wider design. They will be built only if time remains after the core system is finished. The system will not handle money or fundraising, and it will not decide which households are eligible for aid. It also will not support rescue or emergency dispatch, and it will not connect to government or agency systems. In this version, head-of-

household names can be entered in English letters only. The system will be tested with simulated devices and synthetic data, not in a real disaster deployment.

2.5 Feasibility Study

The Offline-First Disaster Relief Distribution & Coordination System gives independent volunteer groups a shared record of relief distribution that still works offline during a disaster. This study asks whether a university capstone team can build and use it. It looks at technical, operational, economic and scheduling feasibility, and at security and privacy.

1. Technical Feasibility
    (a) Hardware and Infrastructure: The field application targets low-cost Android phones running Android 8.0 or later with 2 GB of RAM, which most volunteers already own. The server and coordination dashboard need only a standard cloud or university server. No specialised hardware is required.
    (b) Software and Technical Approach: Local storage on the device, synchronisation with a server and record matching are all established techniques, and existing field data-collection tools already show that offline synchronisation works in practice. The main technical risk is getting synchronisation and conflict handling right. We will reduce it by keeping distribution records append-only and by testing with simulated devices and injected network failures.
    (c) Technical Expertise: Most of the work uses mobile, web and database skills from the Software Engineering curriculum. The team will need to learn synchronisation design and name-similarity matching. Limiting this version to names written in English letters keeps the m atching problem manageable.
2. Operational Feasibility

(a) User Adoption: Volunteer groups join voluntarily, and the shared record is useful only if several groups in the same area use it. We support adoption with a simple Bangla interface, a briefing of about ten minutes and free access for participating organisations. Real-world adoption cannot be measured within the project, so we acknowledge it as a limitation.

(b) Impact on Current Practice: The system replaces informal coordination through social media posts, phone calls and private spreadsheets with one shared record. It does not change how groups decide where to go or whom to help. Coordinators get a new review task for possible duplicates and conflicts, and we will tune the duplicate thresholds so that this queue stays manageable.

3. Economic Feasibility

(a) Development Cost: The system will be built with free and open-source tools, so no software licences are needed. The main costs are the team's time, modest server hosting, and test phones that team members already own.

(b) Operating Cost and Sustainability: The system is meant to be free for volunteer groups. Hosting costs during an operation should be small but not zero, and running it long term would need a host organisation such as an NGO or a university. Finding such a host is outside the scope of the capstone.

4. Scheduling Feasibility

(a) Project Timeline: Development will be finished within the capstone course period, in phases: requirements and design, then offline field work and synchronisation, then conflict and duplicate handling with coverage analysis, and finally testing and evaluation. Planned extensions will be added only if time remains after the core system is complete.

(b) Dependencies: The project does not depend on government approval, third-party integrations or a real disaster happening, because the evaluation uses synthetic data. The main schedule risk is that synchronisation and conflict handling take longer than expected, so we will build and test these components first.

5. Security and Privacy Considerations

(a) Data Protection: The system stores names, phone numbers and vulnerability flags for affected families, and exposure could harm them. We reduce this risk by collecting only the data needed for coordination, encrypting data on the phone and in transit, and masking phone numbers for users outside the registering organisation. We also enforce role-based access on the server and delete personal details 90 days after an operation closes.

(b) Legal Considerations: Bangladesh's personal data protection framework requires explicit consent for processing personal data. The system records a household's consent before registration, and relief can be recorded anonymously when consent is refused. We will review the design against the regulations in force at the time of submission.

The project is technically and economically feasible for a student team, because it uses established techniques, free tools and devices that users already own. Its main risks are the correctness of synchronisation and whether independent groups will adopt the system. We address the first through early development and systematic testing, and we acknowledge the second as a limitation. Evaluation with simulated devices and synthetic data can show that the system works correctly, but it cannot show its impact in a real disaster.

2.6 Project Block Diagram

2.6.1 Block Diagram-1: Field Volunteer

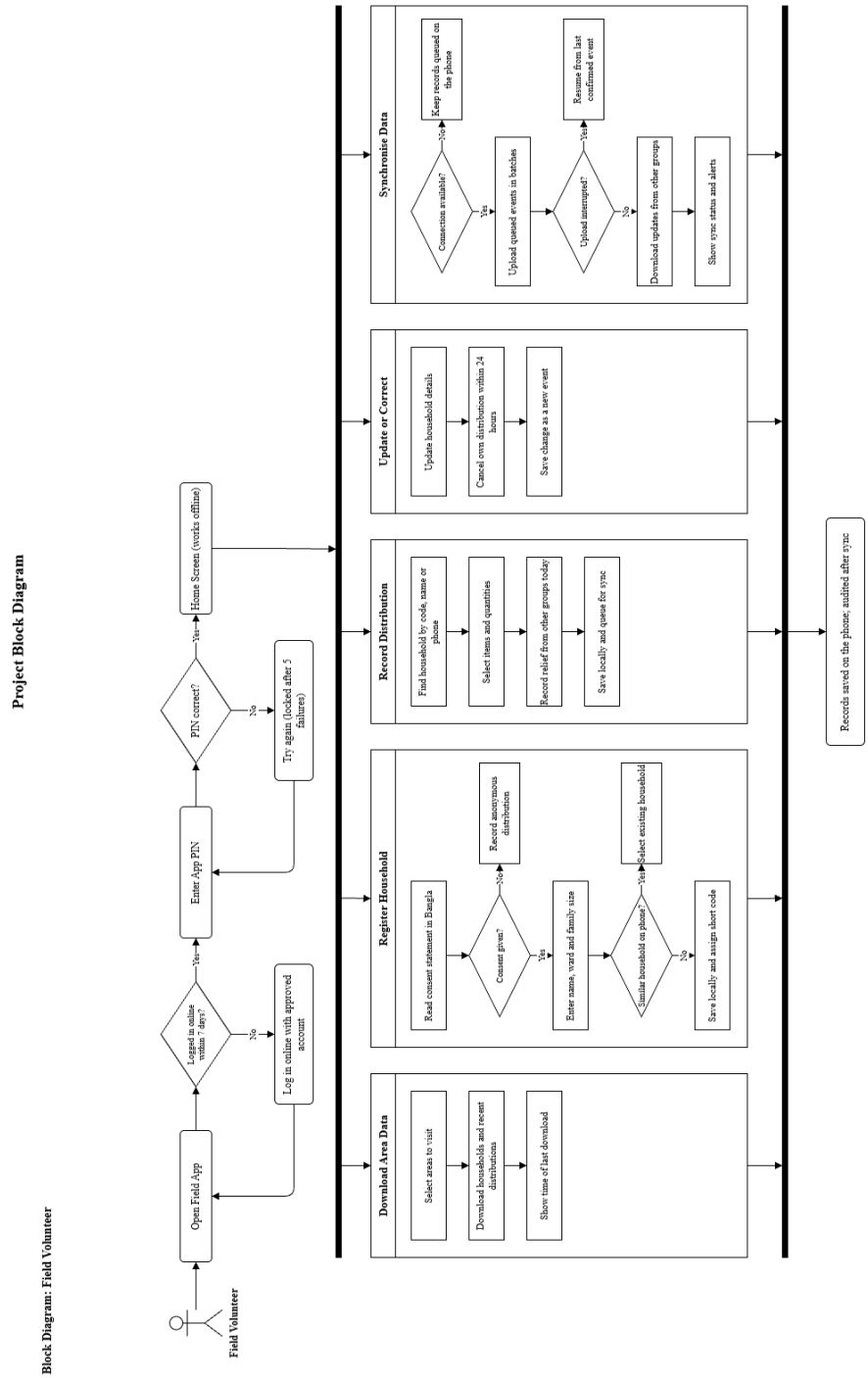


Figure 2.1: Block Diagram for Field Volunteer

2.6.2 Block Diagram-2: Group Coordinator

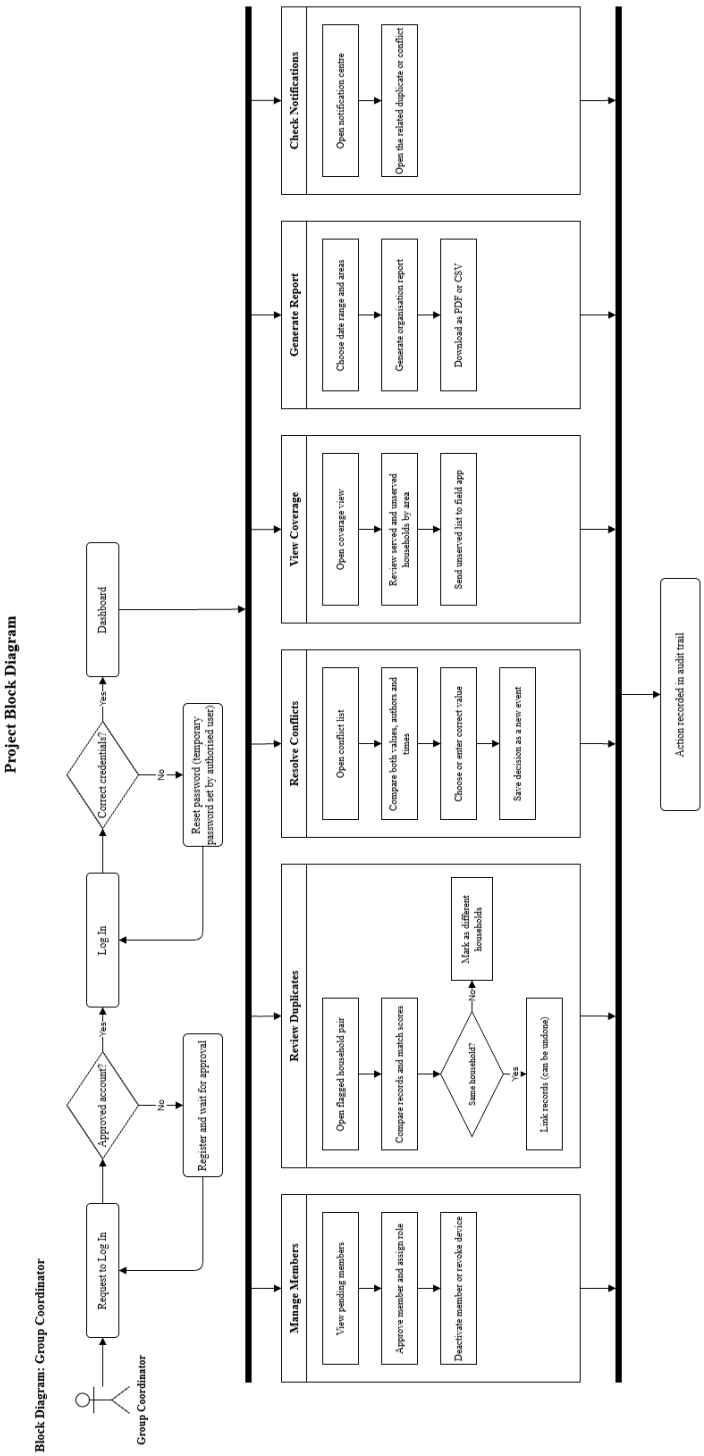


Figure 2.2: Block Diagram for Group Coordinator

2.6.3 Block Diagram-3: Operation Administrator

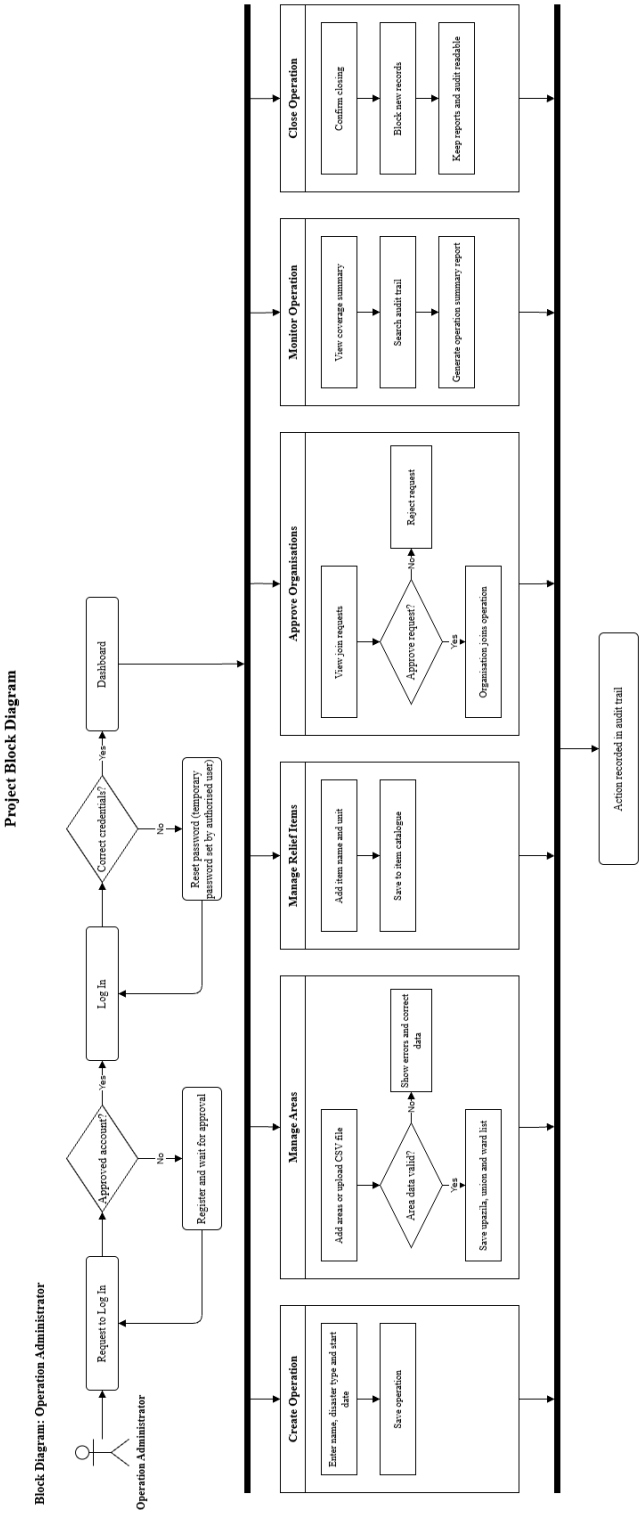


Figure 2.3: Block Diagram for Operation Administrator

2.6.4 Block Diagram-4: System Administrator

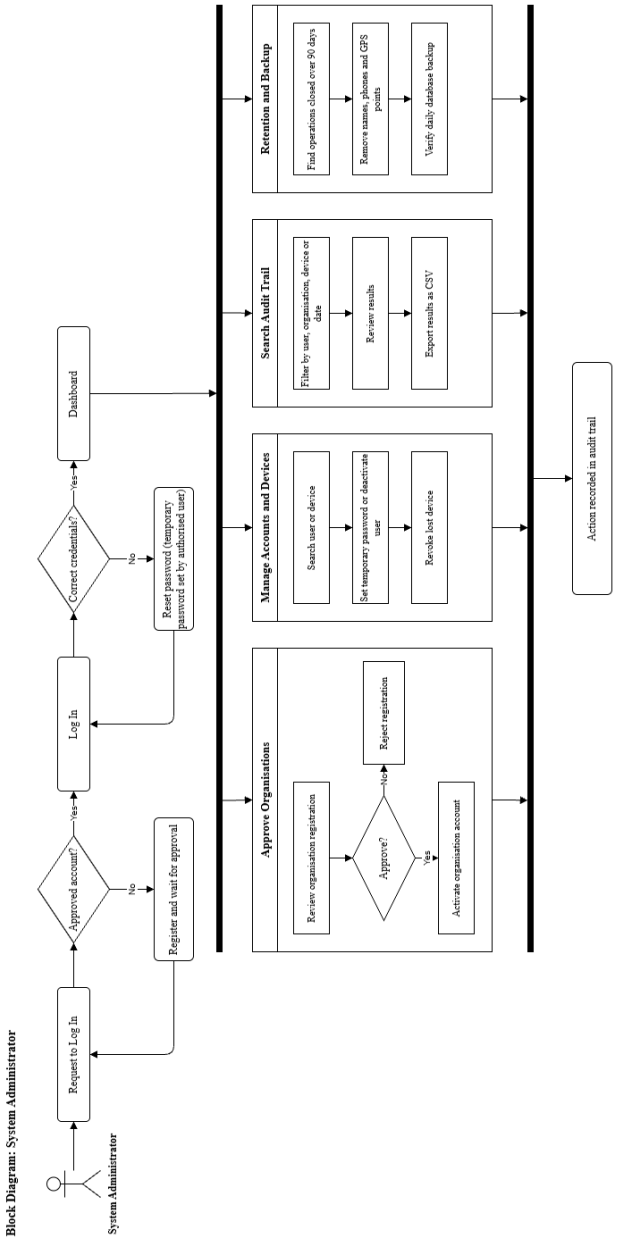


Figure 2.4: Block Diagram for System Administrator

2.6.5 Block Diagram-5: Observer

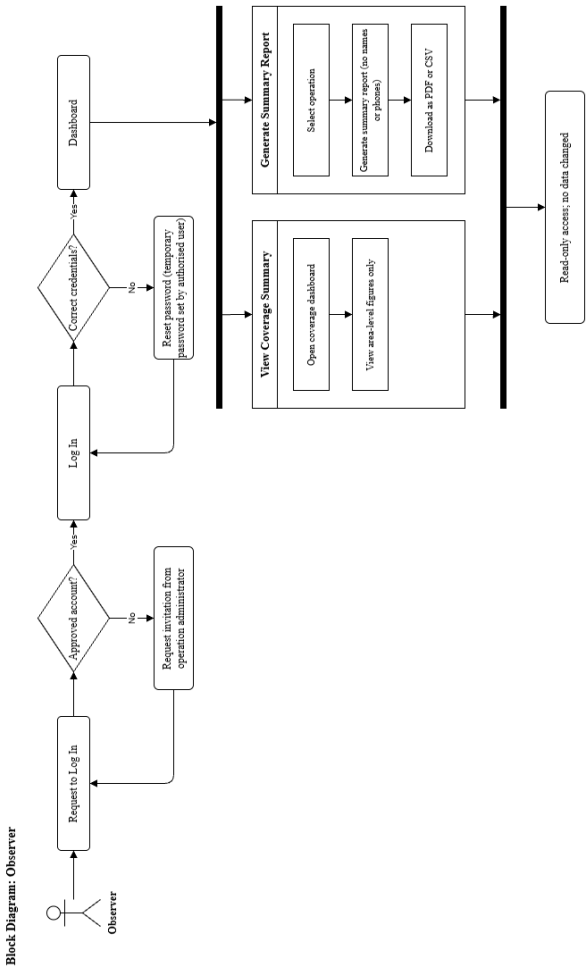


Figure 2.5: Block Diagram for Observer

CHAPTER 3: SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Functional Requirements

FR01	**Registration & Log-In**
Description	A user registers with name, mobile number and password under an organisation, and can log in only after the account is approved.
Stakeholder	Field Volunteer, Group Coordinator, Operation Administrator, System Administrator, Observer

FR02	**Organisation Registration**
Description	A Group Coordinator registers an organisation, and the System Administrator approves or rejects it before it can join an operation.
Stakeholder	Group Coordinator, System Administrator

FR03	**Member and Device Management**
Description	A Group Coordinator approves members, assigns roles, deactivates users and revokes lost devices for their own organisation.
Stakeholder	Group Coordinator, System Administrator

FR04	**Operation Lifecycle Management**
Description	An Operation Administrator creates a disaster operation and closes it when relief work ends, after which no new records are accepted.
Stakeholder	Operation Administrator

FR05	**Area and Relief Item Setup**
Description	The Operation Administrator adds affected areas (upazila, union, ward) manually or by CSV upload, and keeps a list of relief items.
Stakeholder	Operation Administrator, Field Volunteer

FR06	**Join Operation**
Description	A Group Coordinator requests to join an operation, and the Operation Administrator approves or rejects the request.
Stakeholder	Group Coordinator, Operation Administrator

FR07	**Area Data Download**
Description	While online, a volunteer downloads the households and recent distributions of selected areas to the phone for offline use.
Stakeholder	Field Volunteer, Group Coordinator

FR08	**Offline Household Registration**
Description	A volunteer registers a household offline after recording its consent, and the system assigns it a short code.
Stakeholder	Field Volunteer

FR09	**Local Duplicate Warning**
Description	Before saving a new household, the app shows similar households stored on the phone so the volunteer can pick an existing one.
Stakeholder	Field Volunteer

FR10	**Offline Household Search**
Description	A volunteer searches stored households by short code, name or mobile number without an internet connection.
Stakeholder	Field Volunteer, Group Coordinator

FR11	**Relief Distribution Recording**
Description	A volunteer records the items and quantities given to a household offline, or an anonymous distribution if it is not registered.
Stakeholder	Field Volunteer

FR12	**Record Correction and Update**
Description	A volunteer can update household details and cancel their own distribution within 24 hours; cancelled records are kept as voided.
Stakeholder	Field Volunteer, Group Coordinator

FR13	**Data Synchronisation**
Description	When a connection is available, the app syncs automatically or on request and shows the pending records and last sync time.
Stakeholder	Field Volunteer, Group Coordinator

FR14	Resumable and Idempotent Upload
Description	Records are uploaded in batches; an interrupted upload resumes from where it stopped, and a repeated record is stored only once.
Stakeholder	Field Volunteer

FR15	Shared Update Download
Description	During sync, the app downloads new records added by other devices and organisations for the selected areas.
Stakeholder	Field Volunteer, Group Coordinator

FR16	Change Merging and Conflict Detection
Description	The server merges edits to different fields automatically and marks a conflict when the same field is given different values.
Stakeholder	Group Coordinator

FR17	Conflict Resolution
Description	A Group Coordinator compares conflicting values and chooses or enters the correct one, which is saved as a new record.
Stakeholder	Group Coordinator

FR18	Duplicate Household Scoring
Description	After sync, the server compares households in the same union and labels likely or possible duplicates using a similarity score.
Stakeholder	Group Coordinator

FR19	Duplicate Review and Linking
Description	A Group Coordinator decides whether a flagged pair is the same household; confirmed pairs are linked, and the link can be undone.
Stakeholder	Group Coordinator

FR20	Possible Duplicate Distribution Flag
Description	The system flags a household that receives the same item from two organisations within 24 hours for review.
Stakeholder	Group Coordinator, Operation Administrator

FR21	**Area Coverage Summary**
Description	For each area, the dashboard shows served and unserved households along with a coverage status label.
Stakeholder	Group Coordinator, Operation Administrator, Observer

FR22	**Unserved Household List**
Description	A Group Coordinator views the households in an area with no recorded relief and sends the list to the field app.
Stakeholder	Group Coordinator, Field Volunteer

FR23	**In-App Notifications**
Description	Coordinators are notified of new duplicates and conflicts, and volunteers are notified when a sync fails or records wait too long.
Stakeholder	Field Volunteer, Group Coordinator, Operation Administrator

FR24	**Audit Trail Recording**
Description	The system records every registration, distribution, correction, review decision and log-in with the user, organisation, device and time.
Stakeholder	Group Coordinator, Operation Administrator, System Administrator

FR25	**Household History and Audit Search**
Description	Authorised users view a household's full history, and administrators search the audit trail and export results as CSV.
Stakeholder	Group Coordinator, Operation Administrator, System Administrator

FR26	**Reports and Export**
Description	Coordinators generate organisation reports, and administrators and observers generate operation summaries in PDF or CSV.
Stakeholder	Group Coordinator, Operation Administrator, Observer

3.2 Non-Functional Requirements

3.2.1 Security

NFR01	**Role-Based Access Control**
Description	Each user can perform only the actions allowed for their role and organisation, and every request is checked on the server.
Stakeholder	Field Volunteer, Group Coordinator, Operation Administrator, System Administrator, Observer

NFR02	**Authentication and Session Security**
Description	Passwords are stored as salted hashes, accounts lock after 5 failed log-ins, and the offline app is protected by a PIN for up to 7 days.
Stakeholder	Field Volunteer, Group Coordinator, System Administrator

NFR03	**Password Recovery**
Description	A user who forgets their password gets a temporary password from their coordinator or the System Administrator.
Stakeholder	Field Volunteer, Group Coordinator, System Administrator

NFR04	**Data Encryption**
Description	All communication uses HTTPS, and the local database on each phone is encrypted.
Stakeholder	Field Volunteer, System Administrator

3.2.2 Privacy

NFR05	**Data Minimisation and Consent**
Description	The system stores only the household data needed for coordination and registers a household only with its consent.
Stakeholder	Field Volunteer, Operation Administrator

NFR06	**Restricted Data Visibility**
Description	Phone numbers are masked outside the registering organisation, and Observers see only area-level figures.
Stakeholder	Group Coordinator, Operation Administrator, System Administrator, Observer

NFR07	Data Retention
Description	Names, phone numbers and GPS points are removed 90 days after an operation closes.
Stakeholder	Operation Administrator, System Administrator

3.2.3 Reliability

NFR08	Offline Operation and Local Persistence
Description	Field functions work offline for at least 3 days, and no saved record is lost after an app crash, restart or battery failure.
Stakeholder	Field Volunteer, Group Coordinator

NFR09	Exactly-Once Event Storage
Description	Each record is stored on the server exactly once, even when uploads are repeated or arrive out of order.
Stakeholder	Field Volunteer, Group Coordinator

NFR10	Eventual Consistency
Description	After all devices finish syncing, every device and the server show the same data for shared areas.
Stakeholder	Field Volunteer, Group Coordinator

NFR11	Conflict Visibility
Description	No edit is silently overwritten; all same-field conflicts must appear in the review list.
Stakeholder	Group Coordinator

NFR12	Data Integrity and Traceability
Description	Records are append-only and traceable to their user and device, audit logs cannot be changed, and input is validated.
Stakeholder	Group Coordinator, Operation Administrator, System Administrator

NFR13	Backup and Recovery
Description	The server database is backed up daily, and a restore is tested at least once during the project.
Stakeholder	System Administrator

3.2.4 Performance

NFR14	Field Application Performance
Description	The app stores at least 5,000 households and saves or searches records within 1 second on a phone with 2 GB of RAM.
Stakeholder	Field Volunteer

NFR15	Synchronisation Performance
Description	Syncing 500 pending records completes within 60 seconds on a slow (about 400 kbps) connection.
Stakeholder	Field Volunteer

NFR16	Server Performance
Description	Dashboard views load within 3 seconds for 20,000 households, and the server handles 50 devices syncing at once.
Stakeholder	Group Coordinator, Operation Administrator, System Administrator

3.2.5 Accuracy

NFR17	Duplicate Detection Accuracy
Description	On labelled test data, duplicate detection reaches at least 90% precision and 80% recall.
Stakeholder	Group Coordinator

3.2.6 Usability

NFR18	Language Support
Description	The field app is available in Bangla and English, and the dashboard shows coverage status labels in Bangla.
Stakeholder	Field Volunteer, Group Coordinator

NFR19	Field Usability
Description	After a 10-minute briefing, a new volunteer can register a household and record a distribution within 2 minutes using one hand.
Stakeholder	Field Volunteer

3.2.7 Operational

NFR20	**Platform Compatibility**
Description	The field app runs on Android 8.0 or later, and the dashboard works on current Chrome, Firefox and Edge browsers.
Stakeholder	Field Volunteer, Group Coordinator, Operation Administrator

NFR21	**Maintainability**
Description	Disaster types, areas and items are configurable without code changes, and core modules are covered by automated tests.
Stakeholder	System Administrator

CHAPTER 4: USE CASE MODELLING

4.1 Use Case Diagram

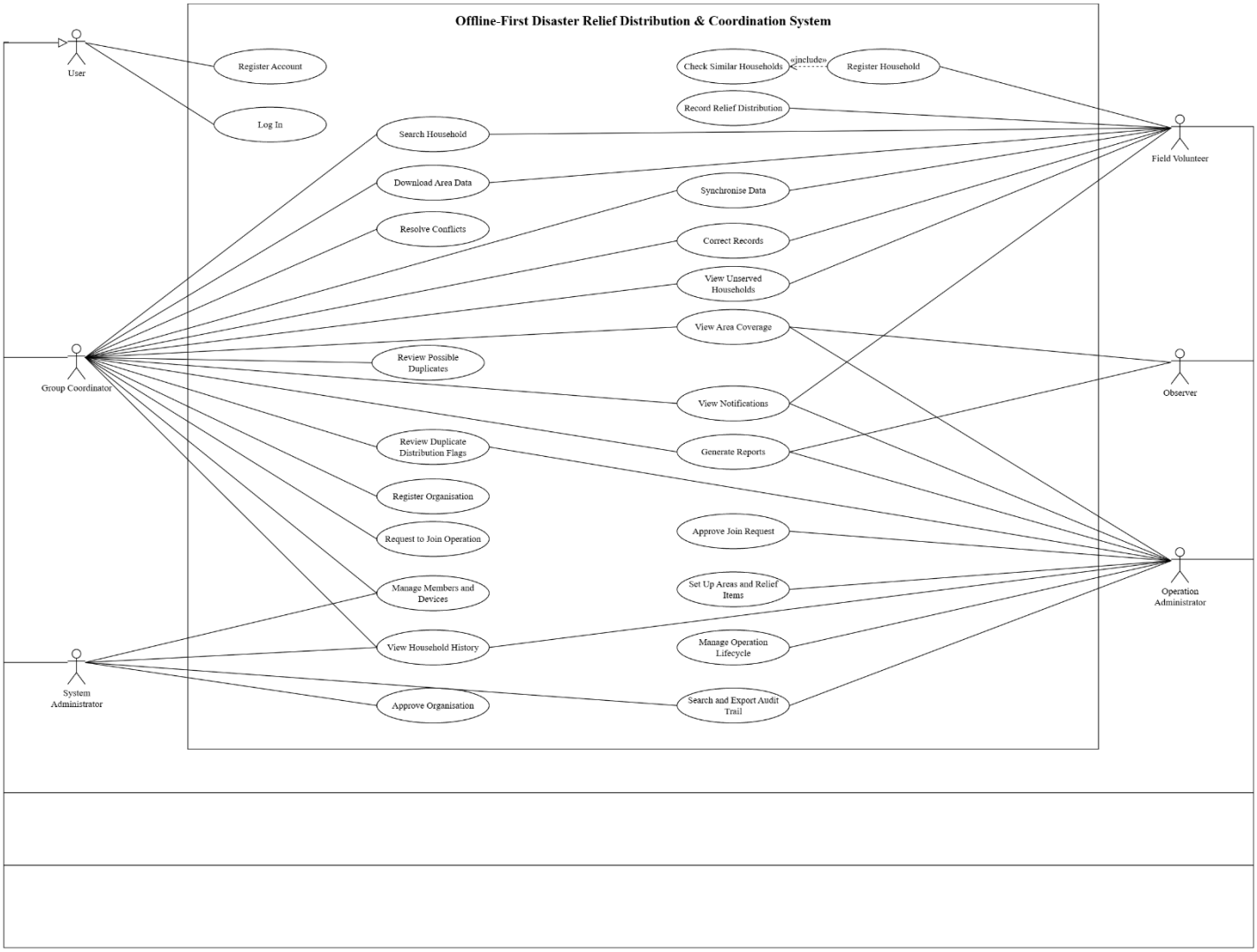


Figure 4.1: Use Case Diagram

4.2 Use Case Description

4.2.1 Use Case Description-01: Register Account

<table>
<tr><td>Use Case</td><td>Register Account</td></tr>
<tr><td>Goal</td><td>A user requests an account under an approved organisation.</td></tr>
<tr><td>Preconditions</td><td>The organisation is approved, and internet access is available for registration.</td></tr>
<tr><td>Success End Condition</td><td>Account saved as pending. Message: 'Registration submitted for approval.'</td></tr>
<tr><td>Failed End Condition</td><td>No account created. Message: 'Registration could not be completed.'</td></tr>
<tr><td>Primary Actors:

Secondary Actors:</td><td>Field Volunteer, Group Coordinator, Operation Administrator, System Administrator, Observer
Group Coordinator, System Administrator</td></tr>
<tr><td>Trigger</td><td>The user selects Register on the field app or dashboard.</td></tr>
<tr><td>Description / Main Success Scenario</td><td>
<table>
<tr><td>1.</td><td>User opens the registration form.</td></tr>
<tr><td>2.</td><td>User enters name, mobile number and password.</td></tr>
<tr><td>3.</td><td>User selects their organisation.</td></tr>
<tr><td>4.</td><td>System validates the details.</td></tr>
<tr><td>5.</td><td>System saves the account as pending approval.</td></tr>
<tr><td>6.</td><td>System shows that the account is waiting for approval.</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flows</td><td>
<table>
<tr><td>1.a.</td><td>The device is offline.</td></tr>
<tr><td></td><td>1.a.1. System asks the user to connect and try again.</td></tr>
<tr><td>2.a.</td><td>The mobile number is already registered.</td></tr>
<tr><td></td><td>2.a.1. System shows an error and asks the user to log in.</td></tr>
<tr><td>4.a.</td><td>A required field is missing or invalid.</td></tr>
<tr><td></td><td>4.a.1. System marks the field and asks for a correction.</td></tr>
</table>
</td></tr>
<tr><td>Quality Requirements</td><td>1. Store passwords using salted bcrypt or Argon2 hashes.
2. Provide Bangla and English forms in the field app.</td></tr>
</table>

4.2.2 Use Case Description-02: Log In

Use Case	Log In	
Goal	An approved user gains access to their authorised functions.	
Preconditions	The account is active, and the device has not been revoked.	
Success End Condition	Access granted. Message: 'Login successful.'	
Failed End Condition	Access denied. Message: 'Unable to log in; check your account details.'	
Primary Actors: Secondary Actors:	Field Volunteer, Group Coordinator, Operation Administrator, System Administrator, Observer None	
Trigger	The user opens the field app or dashboard and selects Log In.	
Description / Main Success Scenario	1.	User enters mobile number and password.
	2.	System checks the credentials on the server.
	3.	System loads the user's role and organisation.
	4.	System opens the home screen for that role.
	5.	Field app saves the session and asks the user to set a PIN.
	6.	System records the log-in in the audit trail.
Alternative Flows	1.a.	The device is offline and a saved session exists.
		1.a.1. User unlocks the field app with the PIN.
		1.a.2. Offline functions open; server functions stay unavailable.
	1.b.	The saved session is older than 7 days.
		1.b.1. Field app asks the user to connect and log in online.
	2.a.	The password is wrong.
		2.a.1. System shows an error and counts the failed attempt.
	2.b.	The user has forgotten the password.
		2.b.1. User gets a temporary password from the coordinator or System Administrator.
	3.a.	The account is pending or deactivated.
		3.a.1. System denies access and shows the account status.
Quality Requirements	1. Lock online login for 15 minutes after 5 failed attempts. 2. Allow PIN access to a valid offline session for up to 7 days.	

4.2.3 Use Case Description-03: Register Organisation

Use Case	Register Organisation
Goal	A coordinator requests registration of a relief organisation.
Preconditions	The organisation is not registered; the registration service is reachable.
Success End Condition	Organisation saved as pending. Message: 'Registration request submitted.'
Failed End Condition	No organisation created. Message: 'Registration request failed.'
Primary Actors: Secondary Actors:	Group Coordinator System Administrator
Trigger	The coordinator selects Register Organisation on the dashboard.
Description / Main Success Scenario	1. Coordinator opens the organisation registration form. 2. Coordinator enters the organisation name and contact details. 3. Coordinator enters their own account details. 4. System validates the details. 5. System saves the organisation as pending. 6. System sends the request to the System Administrator.
Alternative Flows	4.a. An organisation with the same name already exists. 4.a.1. System rejects the request and shows the existing organisation. 4.b. A required field is missing or invalid. 4.b.1. System asks the coordinator to correct it. 6.a. The System Administrator rejects the request. 6.a.1. System marks the organisation as rejected and shows the reason.
Quality Requirements	1. Validate mandatory fields before accepting a registration. 2. Keep new organisations inactive until approved.

4.2.4 Use Case Description-04: Approve Organisation

Use Case	Approve Organisation
Goal	The System Administrator checks a pending organisation and approves or rejects it.
Preconditions	The System Administrator is signed in; an organisation is awaiting review.
Success End Condition	Organisation activated. Message: 'Organisation approved.'
Failed End Condition	No approval saved. Message: 'Approval was not completed.'
Primary Actors: Secondary Actors:	System Administrator Group Coordinator
Trigger	The System Administrator opens the list of pending organisations.
Description / Main Success Scenario	1. System Administrator opens the pending organisation list. 2. System shows the organisation and coordinator details. 3. System Administrator reviews the details. 4. System Administrator selects Approve. 5. System activates the organisation and its coordinator account. 6. System records the decision in the audit trail.
Alternative Flows	4.a. The details are false or incomplete. 4.a.1. System Administrator selects Reject and enters a reason. 4.a.2. System marks the organisation as rejected.
Quality Requirements	1. Limit organisation approval to the System Administrator role. 2. Record the decision and timestamp in the audit trail.

4.2.5 Use Case Description-05: Manage Members and Devices

Use Case	Manage Members and Devices
Goal	Keep the organisation's members, roles and devices under control.
Preconditions	An authorised administrator is signed in; the organisation is approved.
Success End Condition	Changes recorded. Message: 'Member and device details updated.'
Failed End Condition	No changes saved. Message: 'Unable to update member or device.'
Primary Actors: Secondary Actors:	Group Coordinator, System Administrator Field Volunteer
Trigger	The user opens the Members page.
Description / Main Success Scenario	1. User opens the member list. 2. System shows members with role, status and devices. 3. User approves a pending member. 4. User assigns a role to the member. 5. System saves the changes. 6. System records the action in the audit trail.
Alternative Flows	3.a. A member has left the organisation. 3.a.1. User deactivates the member, and log-in is blocked. 3.b. A member's phone is lost. 3.b.1. User revokes the device, and the server rejects its sync. 3.c. A member has forgotten the password. 3.c.1. User issues a temporary password. 3.d. A coordinator selects a member of another organisation. 3.d.1. System denies the action.
Quality Requirements	1. Limit coordinators to members of their own organisation. 2. Reject further server sync from revoked devices.

4.2.6 Use Case Description-06: Manage Operation Lifecycle

Use Case	Manage Operation Lifecycle
Goal	The Operation Administrator opens an operation for a disaster and closes it when work ends.
Preconditions	The Operation Administrator has an active authorised session.
Success End Condition	Operation created or closed. Message: 'Operation status updated.'
Failed End Condition	Status unchanged. Message: 'Operation update failed.'
Primary Actors: Secondary Actors:	Operation Administrator Group Coordinator
Trigger	The Operation Administrator selects Create Operation.
Description / Main Success Scenario	1. Administrator opens the Operations page. 2. Administrator selects Create and enters operation details. 3. System validates the details. 4. System saves the operation as active. 5. System lists the operation for organisations to join.
Alternative Flows	3.a. A required field is missing. 3.a.1. System asks the administrator to complete it. 4.a. Relief work has ended and the administrator selects Close. 4.a.1. System asks for confirmation. 4.a.2. System closes the operation and stops accepting new records.
Quality Requirements	1. Store disaster types as configurable operation data. 2. Remove personal identifiers 90 days after closure.

4.2.7 Use Case Description-07: Set Up Areas and Relief Items

Use Case	Set Up Areas and Relief Items
Goal	Define the affected areas and relief items of an operation.
Preconditions	The Operation Administrator is signed in; an operation is active.
Success End Condition	Areas and items saved. Message: 'Operation setup saved.'
Failed End Condition	No changes saved. Message: 'Setup could not be saved.'
Primary Actors: Secondary Actors:	Operation Administrator Field Volunteer
Trigger	The administrator opens the setup page of the operation.
Description / Main Success Scenario	1. Administrator opens the setup page. 2. Administrator adds areas by upazila, union and ward. 3. Administrator adds relief items. 4. System validates the entries. 5. System saves the areas and items. 6. System makes the areas available for download.
Alternative Flows	2.a. The administrator has an area list in a CSV file. 2.a.1. Administrator uploads the CSV file. 2.a.2. System imports valid rows and lists rejected rows. 4.a. The area already exists in the same union. 4.a.1. System rejects the duplicate entry.
Quality Requirements	1. Allow areas and relief items to be configured without code changes. 2. Reject invalid or repeated area entries.

4.2.8 Use Case Description-08: Request to Join Operation

Use Case	Request to Join Operation
Goal	A Group Coordinator asks to bring their organisation into an active operation.
Preconditions	An approved coordinator is signed in; the selected operation is active.
Success End Condition	Join request saved as pending. Message: 'Request submitted.'
Failed End Condition	No request saved. Message: 'Join request was not submitted.'
Primary Actors: Secondary Actors:	Group Coordinator Operation Administrator
Trigger	The coordinator selects Join on an active operation.
Description / Main Success Scenario	1. Coordinator views the list of active operations. 2. Coordinator selects an operation. 3. Coordinator sends a join request. 4. System saves the request as pending. 5. System shows the request status.
Alternative Flows	3.a. The organisation has already joined or has a pending request. 3.a.1. System shows the current status and blocks a second request.
Quality Requirements	1. Reject duplicate join requests for the same operation. 2. Accept requests only from approved organisations.

4.2.9 Use Case Description-09: Approve Join Request

Use Case	Approve Join Request
Goal	The Operation Administrator decides which organisations take part in an operation.
Preconditions	The Operation Administrator is signed in; a join request is pending.
Success End Condition	Organisation admitted. Message: 'Join request approved.'
Failed End Condition	Approval not completed. Message: 'Request not approved.'
Primary Actors: Secondary Actors:	Operation Administrator Group Coordinator
Trigger	The administrator opens the join request list.
Description / Main Success Scenario	1. Administrator opens the join request list. 2. System shows the organisation details. 3. Administrator selects Approve. 4. System adds the organisation to the operation. 5. System records the decision in the audit trail.
Alternative Flows	3.a. The organisation is not suitable for this operation. 3.a.1. Administrator rejects the request, and the coordinator sees the result.
Quality Requirements	1. Restrict approval to the Operation Administrator. 2. Record each decision and timestamp in the audit trail.

4.2.10 Use Case Description-10: Download Area Data

Use Case	Download Area Data
Goal	Store the households and recent distributions of selected areas on the phone for offline work.
Preconditions	An authorised device is online; its organisation belongs to the operation.
Success End Condition	Area data stored locally. Message: 'Area download complete.'
Failed End Condition	Old data retained. Message: 'Download incomplete; please retry.'
Primary Actors: Secondary Actors:	Field Volunteer, Group Coordinator None
Trigger	The user selects Download Areas on the field app.
Description / Main Success Scenario	1. User opens the area list. 2. User selects one or more areas. 3. System checks the user's permission. 4. Server sends households, recent distributions and relief items. 5. Field app saves the data in the local database. 6. Field app shows the download time and record count.
Alternative Flows	4.a. The connection drops during download. 4.a.1. Field app keeps the old data and offers a retry. 5.a. Phone storage is low. 5.a.1. Field app warns the user and stops the download.
Quality Requirements	1. Encrypt downloaded data in the local database. 2. Support at least 5,000 locally stored households.

4.2.11 Use Case Description-11: Register Household

Use Case	Register Household
Goal	Record a new affected household on the phone, even without internet.
Preconditions	The volunteer has an authorised session and previously downloaded area data.
Success End Condition	Household saved and queued. Message: 'Household registered; sync pending.'
Failed End Condition	No record saved. Message: 'Household registration failed.'
Primary Actors: Secondary Actors:	Field Volunteer None
Trigger	The volunteer selects New Household.
Description / Main Success Scenario	1. Volunteer selects New Household. 2. Volunteer records the household's consent. 3. Volunteer enters head name, mobile number, members and area. 4. Use Case 'Check Similar Households'. 5. Volunteer confirms it is a new household. 6. Field app saves the household and gives a short code. 7. Field app adds the record to the sync queue.
Alternative Flows	2.a. The household does not give consent. 2.a.1. Registration stops; the volunteer may record an anonymous distribution. 3.a. A required field is missing. 3.a.1. Field app asks the volunteer to complete it. 5.a. The volunteer selects an existing household. 5.a.1. Field app opens that record and saves no new household.
Quality Requirements	1. Save the record locally within 1 second for the target device. 2. Keep saved records after app crashes or restarts.

4.2.12 Use Case Description-12: Check Similar Households

Use Case	Check Similar Households
Goal	Warn the volunteer about similar households on the phone before a new one is saved.
Preconditions	Household registration is underway, with identifying details entered.
Success End Condition	Local comparison completed. Message: 'Similarity check complete.'
Failed End Condition	Comparison unavailable. Message: 'Unable to check similar households.'
Primary Actors: Secondary Actors:	Field Volunteer None
Trigger	Register Household reaches the save step.
Description / Main Success Scenario	1. Field app reads the entered name, mobile number and area. 2. Field app compares them with households on the phone. 3. Field app shows possible local matches, if any. 4. Volunteer reviews matches or continues as a new household.
Alternative Flows	2.a. No similar household is found. 2.a.1. Field app returns to registration. 2.b. A match exists only on another device that has not synced. 2.b.1. It is not shown here; the server checks it after sync.
Quality Requirements	1. Complete local matching within 1 second for 5,000 households.

4.2.13 Use Case Description-13: Record Relief Distribution

Use Case	Record Relief Distribution
Goal	Record the relief items given to a household, even without internet.
Preconditions	The volunteer has an authorised session and the relevant relief-item list.
Success End Condition	Distribution queued. Message: 'Distribution saved; sync pending.'
Failed End Condition	No distribution saved. Message: 'Unable to save distribution.'
Primary Actors: Secondary Actors:	Field Volunteer None
Trigger	The volunteer selects Record Distribution.
Description / Main Success Scenario	1. Volunteer finds the household by short code or name. 2. Field app shows the household and its past distributions. 3. Volunteer selects items and quantities. 4. Volunteer confirms the distribution. 5. Field app saves it with time, user and device. 6. Field app adds it to the sync queue.
Alternative Flows	1.a. The household is not registered or gave no consent. 1.a.1. Volunteer records an anonymous distribution with the area only. 3.a. A quantity is zero or invalid. 3.a.1. Field app asks for a valid quantity.
Quality Requirements	1. Save the distribution locally within 1 second. 2. Keep saved records after app crashes or restarts.

4.2.14 Use Case Description-14: Search Household

Use Case	Search Household
Goal	Find a stored household on the phone without internet.
Preconditions	The user has an authorised session and downloaded household data.
Success End Condition	Matching records displayed. Message: 'Search complete.'
Failed End Condition	No match found. Message: 'No household matches your search.'
Primary Actors: Secondary Actors:	Field Volunteer, Group Coordinator None
Trigger	The user enters a search term on the field app.
Description / Main Success Scenario	1. User opens the search screen. 2. User enters a short code, name or mobile number. 3. Field app searches the local database. 4. Field app shows the results. 5. User opens a household.
Alternative Flows	4.a. No household matches. 4.a.1. Field app suggests checking the spelling or registering a new household. 4.b. The household is in an area not downloaded. 4.b.1. It is not found until that area is downloaded.
Quality Requirements	1. Return local search results within 1 second for 5,000 households. 2. Support Bangla and English text searches.

4.2.15 Use Case Description-15: Correct Records

Use Case	Correct Records
Goal	Fix wrong household details or cancel a wrong distribution.
Preconditions	The user has an authorised session and permission to edit the local record.
Success End Condition	Correction queued. Message: 'Changes saved; sync pending.'
Failed End Condition	Original record retained. Message: 'Changes could not be saved.'
Primary Actors: Secondary Actors:	Field Volunteer, Group Coordinator None
Trigger	The user selects Edit on a household or Cancel on a distribution.
Description / Main Success Scenario	1. User opens the record. 2. User selects Edit. 3. User changes the wrong fields. 4. User saves the change. 5. Field app saves a new version and keeps the old one. 6. Field app adds the change to the sync queue.
Alternative Flows	2.a. The user cancels their own distribution within 24 hours. 2.a.1. Field app marks it as voided and keeps the record. 2.b. More than 24 hours have passed or the record is not the user's. 2.b.1. Field app blocks the cancellation.
Quality Requirements	1. Preserve prior record versions for auditability. 2. Keep pending corrections after app restart.

4.2.16 Use Case Description-16: Synchronise Data

Use Case	Synchronise Data
Goal	Send pending records to the server and receive shared updates from other devices.
Preconditions	An authorised, unrevoked field device has network connectivity.
Success End Condition	Msg: 'Synchronization completed.'
Failed End Condition	Msg: 'Sync interrupted; retry later.'
Primary Actors: Secondary Actors:	Field Volunteer, Group Coordinator None
Trigger	A connection becomes available, or the user taps Sync Now.
Description / Main Success Scenario	1. Field app detects a connection or the user taps Sync Now. 2. Field app sends pending records in batches. 3. Server stores each record once by its unique ID. 4. Server confirms each batch. 5. Server merges incoming changes and checks for conflicts or duplicates. 6. Field app downloads authorised shared updates. 7. Field app removes confirmed items from the queue and shows sync status.
Alternative Flows	2.a. The connection drops during upload. 2.a.1. Field app resumes from the last confirmed batch later. 3.a. A record has already been received. 3.a.1. Server ignores the copy and confirms it. 3.b. The session has expired or the device is revoked. 3.b.1. Server rejects the sync and the app asks the user to log in. 6.a. Sync keeps failing or records wait too long. 6.a.1. Field app notifies the volunteer.
Quality Requirements	1. Synchronize 500 events within 60 seconds at approximately 400 kbps. 2. Store each event once despite retries; retain unconfirmed events.

4.2.17 Use Case Description-17: Resolve Conflicts

Use Case	Resolve Conflicts
Goal	Choose the correct value when two users changed the same field differently.
Preconditions	An authorised coordinator is signed in, and a sync conflict is pending.
Success End Condition	Msg: 'Conflict resolved.'
Failed End Condition	Msg: 'Resolution was not saved.'
Primary Actors: Secondary Actors:	Group Coordinator Field Volunteer
Trigger	The coordinator opens the conflict list or a conflict notification.
Description / Main Success Scenario	1. Coordinator opens the conflict list. 2. Coordinator selects a conflict. 3. System shows both values with user, device and time. 4. Coordinator chooses one value or enters the correct one. 5. System saves it as a new version. 6. System records the decision in the audit trail.
Alternative Flows	4.a. The coordinator needs to check with the field team. 4.a.1. Coordinator leaves the conflict open for later.
Quality Requirements	1. Detect all injected same-field conflicts in controlled tests. 2. Never silently overwrite conflicting values.

4.2.18 Use Case Description-18: Review Possible Duplicates

Use Case	Review Possible Duplicates
Goal	Decide whether two flagged records are the same household.
Preconditions	An authorised coordinator is signed in, and a duplicate candidate exists.
Success End Condition	Msg: 'Duplicate review completed.'
Failed End Condition	Msg: 'Review was not completed.'
Primary Actors: Secondary Actors:	Group Coordinator None
Trigger	The coordinator opens the duplicate list or a duplicate notification.
Description / Main Success Scenario	1. Coordinator opens the duplicate list sorted by score. 2. Coordinator selects a pair. 3. System shows both records side by side with the label. 4. Coordinator confirms they are the same household. 5. System links the records and keeps both. 6. System records the decision in the audit trail.
Alternative Flows	4.a. The records are different households. 4.a.1. Coordinator marks them as not a duplicate. 5.a. The link was made by mistake. 5.a.1. Coordinator undoes the link and the records separate again.
Quality Requirements	1. Target 90% precision and 80% recall on labelled synthetic data. 2. List new duplicate candidates within 2 minutes after sync.

4.2.19 Use Case Description-19: Review Duplicate Distribution Flags

Use Case	Review Duplicate Distribution Flags
Goal	Check households that got the same item from two organisations within 24 hours.
Preconditions	An authorised reviewer is signed in, and a distribution flag is pending.
Success End Condition	Msg: 'Flag review saved.'
Failed End Condition	Msg: 'Flag review was not saved.'
Primary Actors: Secondary Actors:	Group Coordinator, Operation Administrator None
Trigger	The user opens the flag list or a flag notification.
Description / Main Success Scenario	1. User opens the flag list. 2. User selects a flag. 3. System shows both distributions with organisation, item and time. 4. User checks the details. 5. User marks the flag as justified or as a repeat. 6. System saves the decision with a note in the audit trail.
Alternative Flows	4.a. The household itself is waiting in duplicate review. 4.a.1. User reviews the household pair first.
Quality Requirements	Keep flags for human review; never delete distributions automatically.

4.2.20 Use Case Description-20: View Notifications

Use Case	View Notifications
Goal	Users see events that need their attention.
Preconditions	The user has a valid session, and at least one notification is available.
Success End Condition	Notification marked as read. Message: 'Notification opened.'
Failed End Condition	Notification remains unread. Message: 'Unable to open notification.'
Primary Actors: Secondary Actors:	Field Volunteer, Group Coordinator, Operation Administrator None
Trigger	A new notification arrives or the user opens the notification panel.
Description / Main Success Scenario	1. System creates a notification when an event occurs. 2. System shows a badge on the notification icon. 3. User opens the notification list. 4. User selects a notification. 5. System opens the related page. 6. System marks the notification as read.
Alternative Flows	1.a. The volunteer's phone is offline. 1.a.1. Server notifications appear after the next sync.
Quality Requirements	Display notifications only to authorised accounts and organisations.

4.2.21 Use Case Description-21: View Unserved Households

Use Case	View Unserved Households
Goal	Find households with no recorded relief and send the list to the field.
Preconditions	A coordinator is signed in, and synchronized area records are available.
Success End Condition	Msg: 'Unserved list ready.'
Failed End Condition	Msg: 'Unable to prepare the unserved list.'
Primary Actors: Secondary Actors:	Group Coordinator, Field Volunteer None
Trigger	The coordinator selects an area and opens Unserved Households.
Description / Main Success Scenario	1. Coordinator selects an area. 2. System lists households with no recorded distribution. 3. Coordinator reviews the list. 4. Coordinator releases the list to authorised field devices. 5. Volunteer receives the list at the next sync. 6. Volunteer views the list offline.
Alternative Flows	2.a. Every household in the area has been served. 2.a.1. System shows that no household is unserved. 5.a. The volunteer's phone is offline. 5.a.1. The list arrives at the next sync.
Quality Requirements	1. Use synchronized records and display the latest update time. 2. Load area lists within 3 seconds for 20,000 households.

4.2.22 Use Case Description-22: View Area Coverage

Use Case	View Area Coverage
Goal	See how many households in each area have been served.
Preconditions	An authorised user is signed in; synchronized operation records are available.
Success End Condition	Area counts displayed. Message: 'Coverage summary loaded.'
Failed End Condition	No summary displayed. Message: 'Coverage data unavailable.'
Primary Actors: Secondary Actors:	Group Coordinator, Operation Administrator, Observer None
Trigger	The user opens the coverage dashboard.
Description / Main Success Scenario	1. User opens the coverage dashboard. 2. User selects an operation or area. 3. System counts served and unserved households. 4. System shows the summary with a coverage label. 5. User filters by upazila, union or ward.
Alternative Flows	3.a. The user is an Observer. 3.a.1. System shows area-level figures only. 3.b. Some devices have not synced yet. 3.b.1. System shows the last updated time with the figures.
Quality Requirements	1. Load coverage summaries within 3 seconds for 20,000 households. 2. Show coverage-status labels in Bangla.

4.2.23 Use Case Description-23: View Household History

Use Case	View Household History
Goal	See every recorded action on a household in order.
Preconditions	An authorised user is signed in and can access the household record.
Success End Condition	Household activity displayed. Message: 'History loaded.'
Failed End Condition	No history displayed. Message: 'Household history unavailable.'
Primary Actors: Secondary Actors:	Group Coordinator, Operation Administrator, System Administrator None
Trigger	The user opens a household and selects History.
Description / Main Success Scenario	1. User searches for the household. 2. User selects History. 3. System lists registration, distributions, corrections and review decisions. 4. System shows the user, organisation and time of each entry.
Alternative Flows	3.a. The household belongs to another organisation. 3.a.1. System masks the phone number.
Quality Requirements	1. Preserve the append-only household event history. 2. Mask restricted household identifiers for other organisations.

4.2.24 Use Case Description-24: Search and Export Audit Trail

Use Case	Search and Export Audit Trail
Goal	Administrators trace who did what, when and from which device.
Preconditions	An authorised administrator is signed in; audit entries are available.
Success End Condition	Matching audit entries exported. Message: 'Audit export ready.'
Failed End Condition	No export created. Message: 'Audit search or export failed.'
Primary Actors: Secondary Actors:	Operation Administrator, System Administrator None
Trigger	The administrator opens the audit trail page.
Description / Main Success Scenario	1. Administrator opens the audit trail page. 2. Administrator filters by user, organisation, device, date or action. 3. System shows the matching entries. 4. Administrator selects Export CSV. 5. System creates the CSV file for download. 6. System records the export in the audit trail.
Alternative Flows	3.a. No entry matches the filters. 3.a.1. System shows an empty result message.
Quality Requirements	1. Prevent editing or deletion of audit entries through the interface. 2. Restrict audit search and export to authorised administrators.

4.2.25 Use Case Description-25: Generate Reports

Use Case	Generate Reports
Goal	Produce a summary report of an organisation or operation.
Preconditions	An authorised user is signed in; synchronized report data is available.
Success End Condition	Report produced. Message: 'Report ready to download.'
Failed End Condition	No report produced. Message: 'Unable to generate report.'
Primary Actors: Secondary Actors:	Group Coordinator, Operation Administrator, Observer None
Trigger	The user selects Reports on the dashboard.
Description / Main Success Scenario	1. User opens the Reports page. 2. User selects report type and date range. 3. User chooses PDF or CSV. 4. System builds the report from server data. 5. User downloads the report.
Alternative Flows	2.a. The user is a Group Coordinator. 2.a.1. System offers reports for their own organisation only. 2.b. The user is an Observer. 2.b.1. System offers area-level summaries without personal data. 4.a. No data exists for the chosen range. 4.a.1. System shows that the report is empty.
Quality Requirements	1. Use synchronized records and display report generation time. 2. Exclude personal details from Observer reports.

CHAPTER 5: SYSTEM DESIGN

5.1 Data Flow Diagram

5.1.1 Data Flow Diagram Level-0

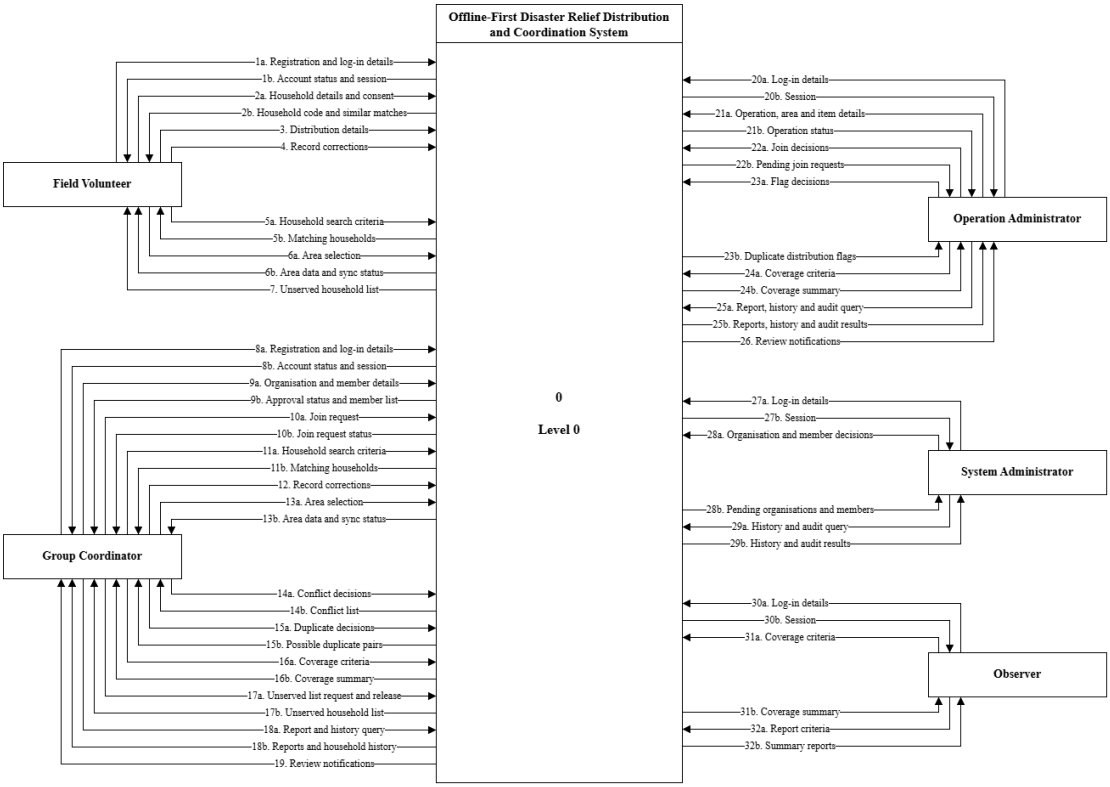


Figure 5.1: Data Flow Diagram Level-0

5.1.2 Data Flow Diagram Level-1

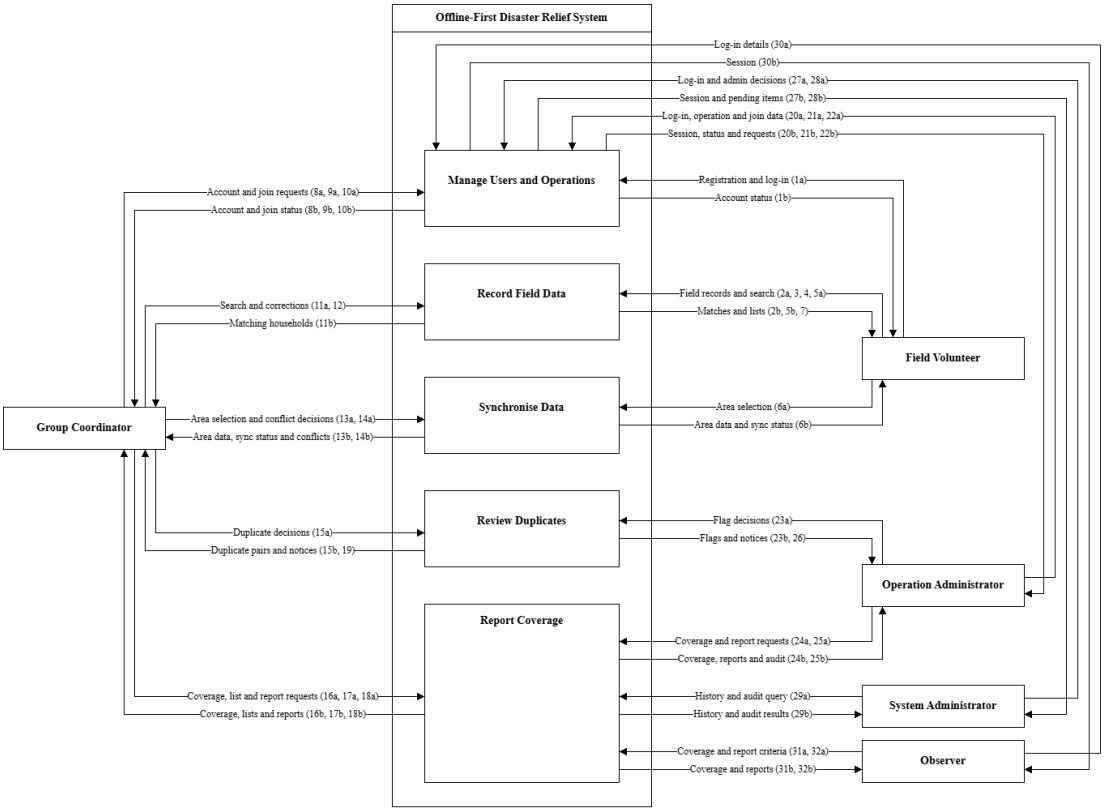


Figure 5.2: Data Flow Diagram Level-1

5.2 Sequence Diagram

5.2.1 Sequence Diagram-01: Register Household Offline

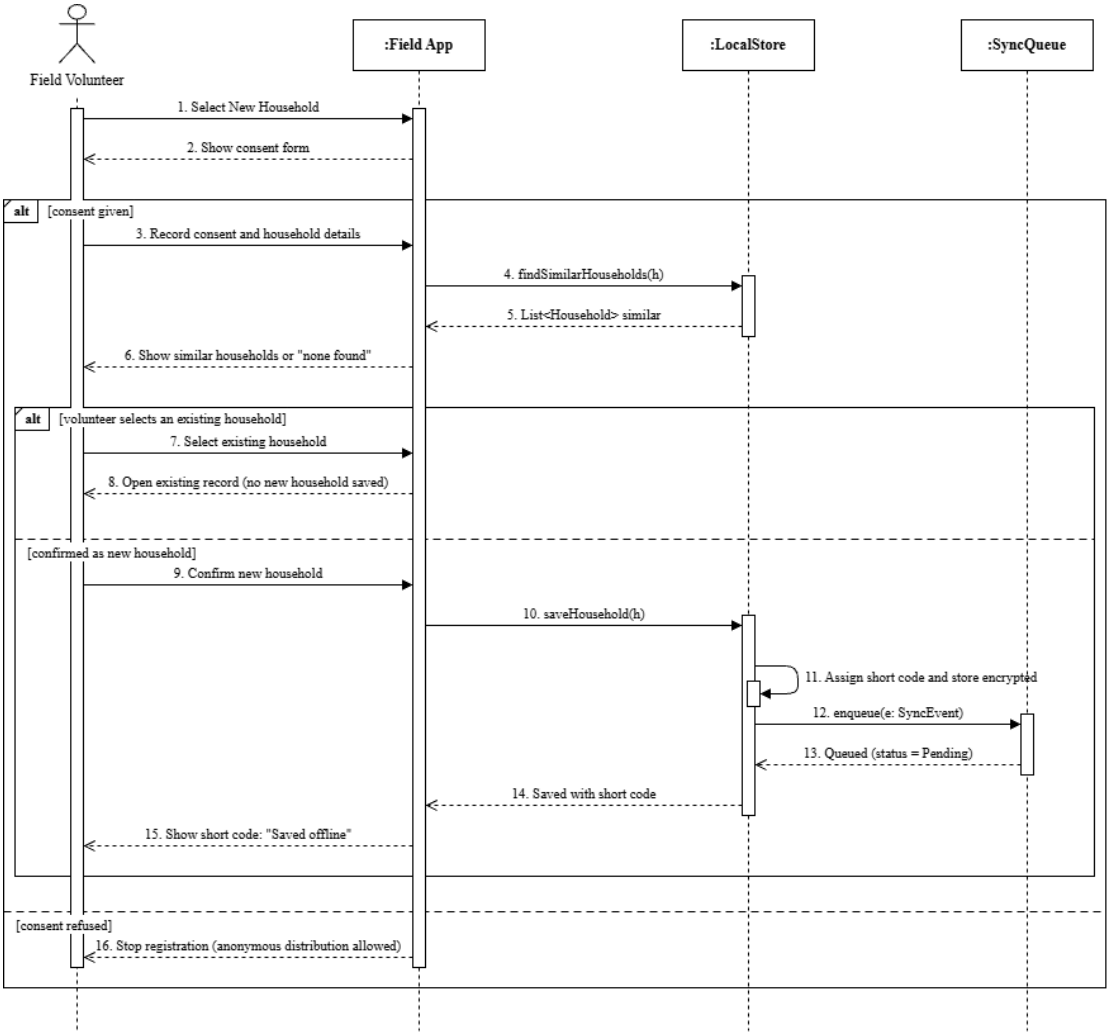


Figure 5.3: Sequence Diagram for Register Household Offline

5.2.2 Sequence Diagram-02: Record Relief Distribution Offline

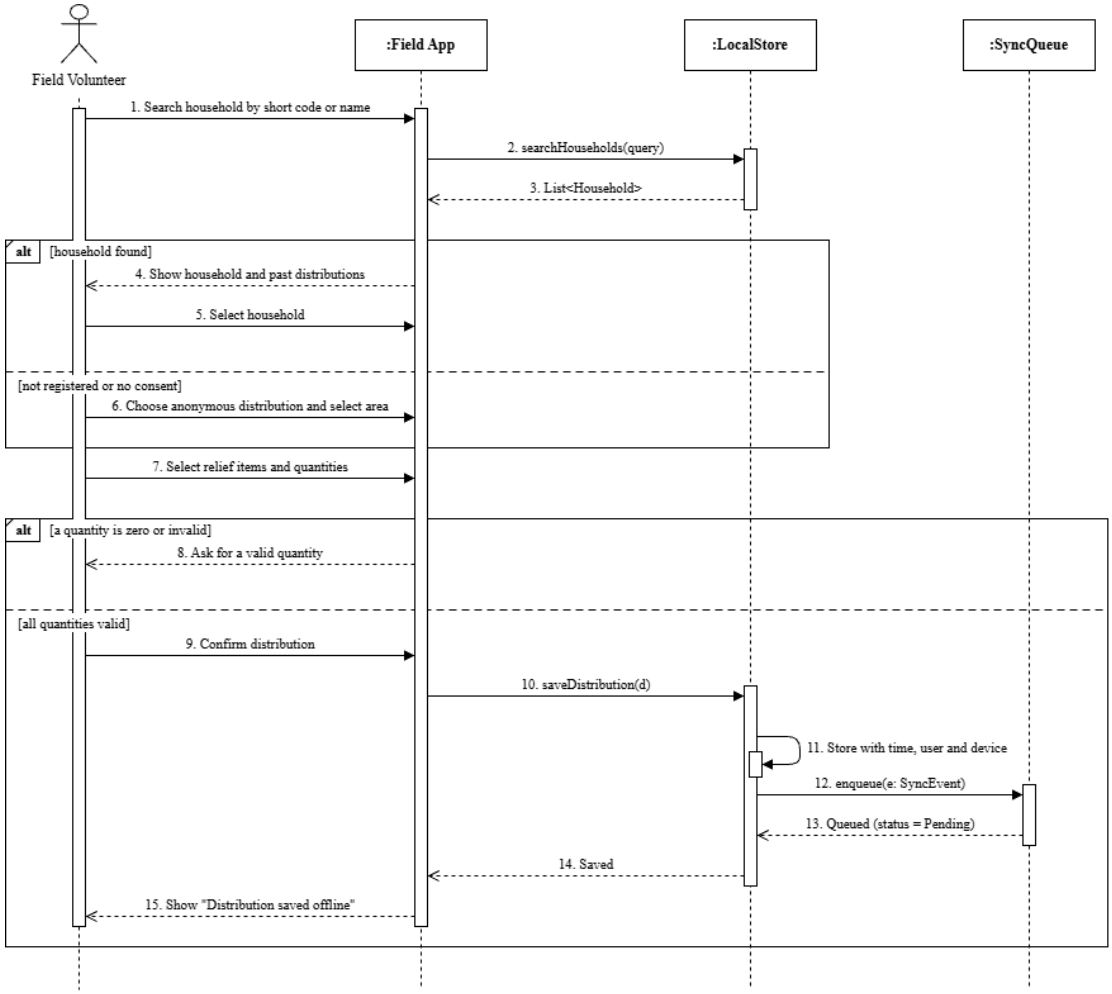


Figure 5.4: Sequence Diagram for Record Relief Distribution Offline

5.2.3 Sequence Diagram-03: Synchronise Data

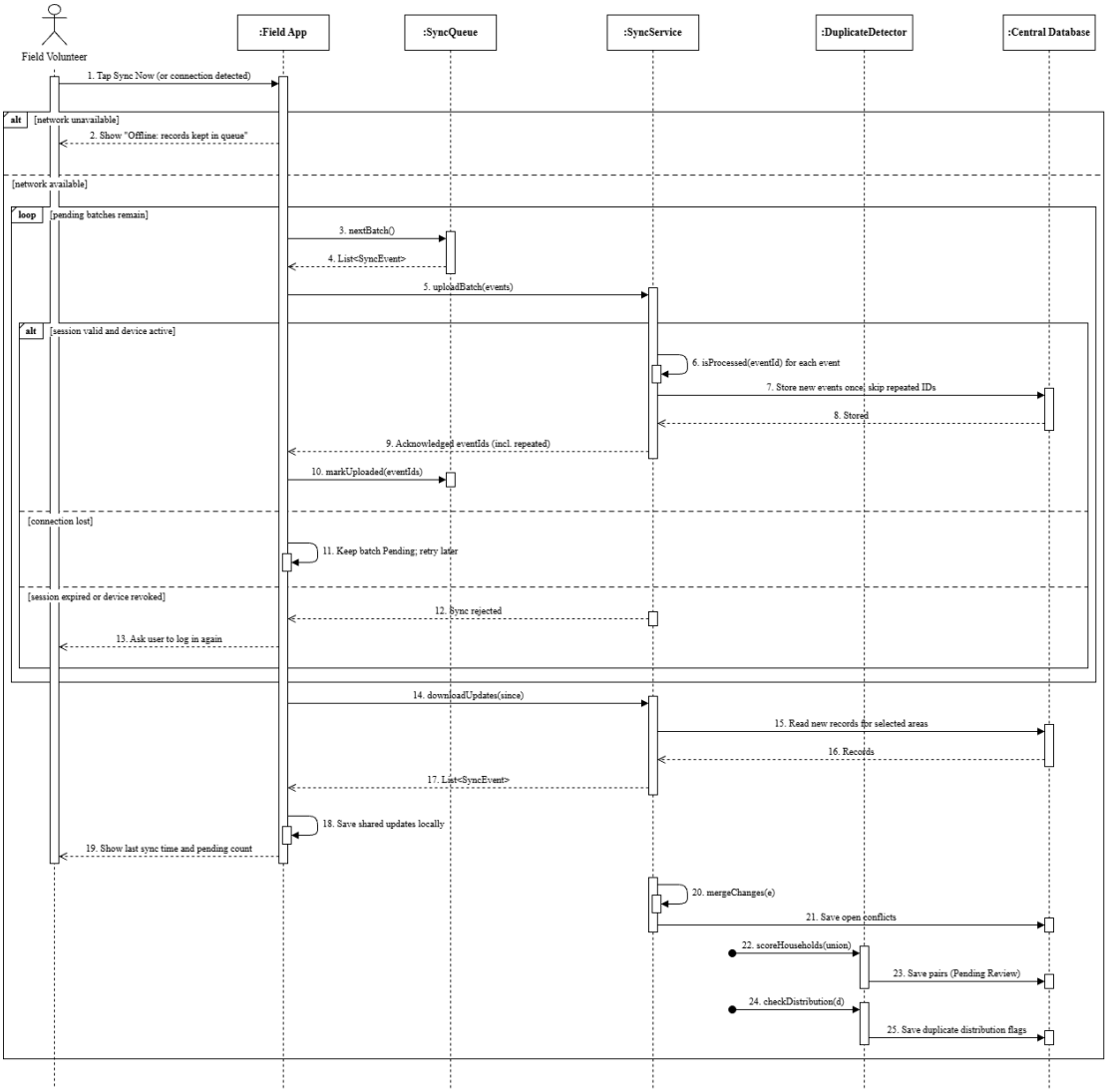


Figure 5.5: Sequence Diagram for Synchronise Data

5.2.4 Sequence Diagram-04: Resolve Conflict

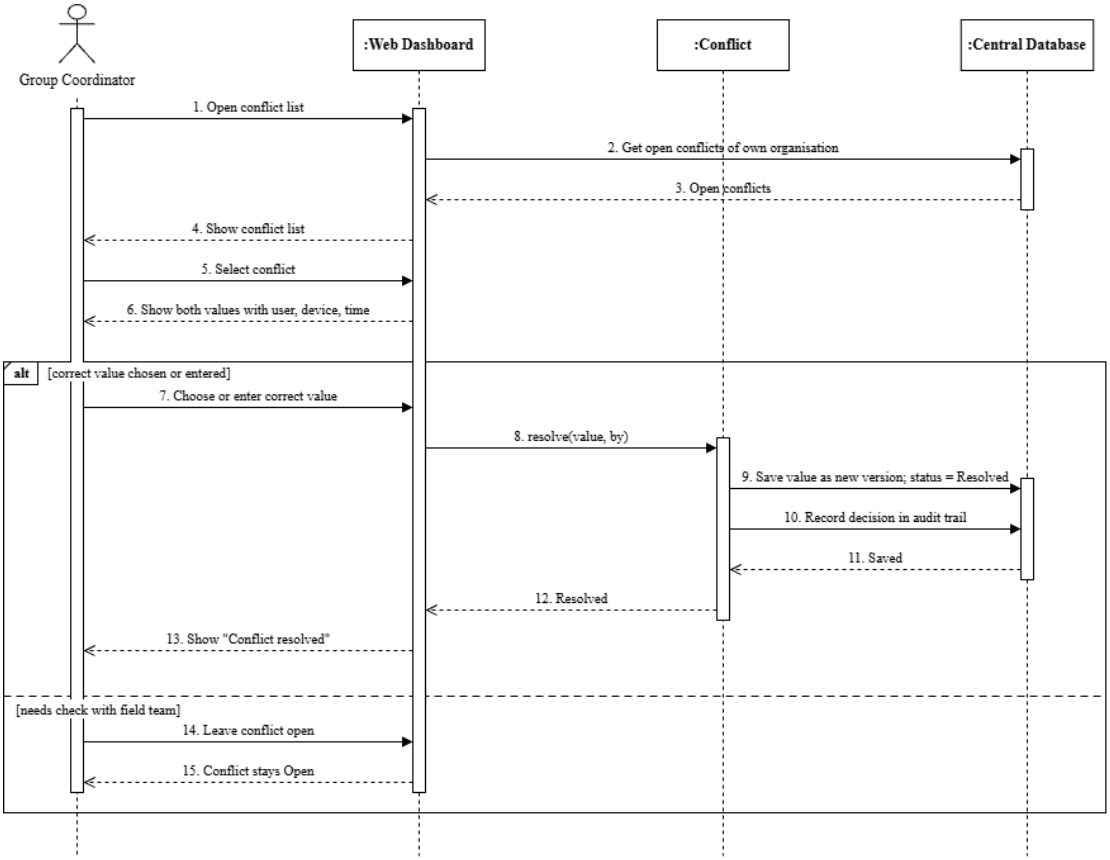


Figure 5.6: Sequence Diagram for Resolve Conflict

5.2.5 Sequence Diagram-05: Review Possible Duplicates

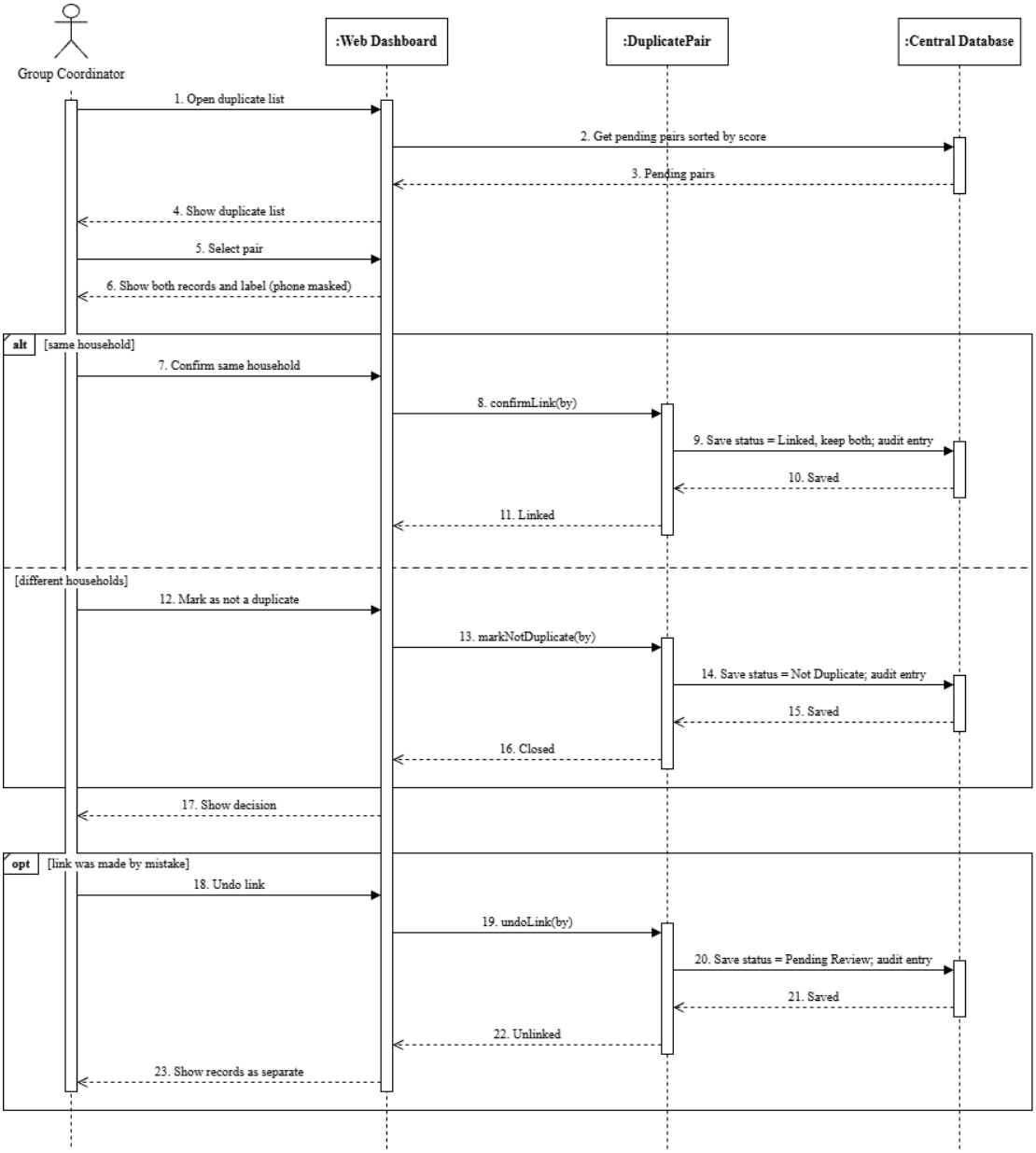


Figure 5.7: Sequence Diagram for Review Possible Duplicates

5.2.6 Sequence Diagram-06: View Area Coverage

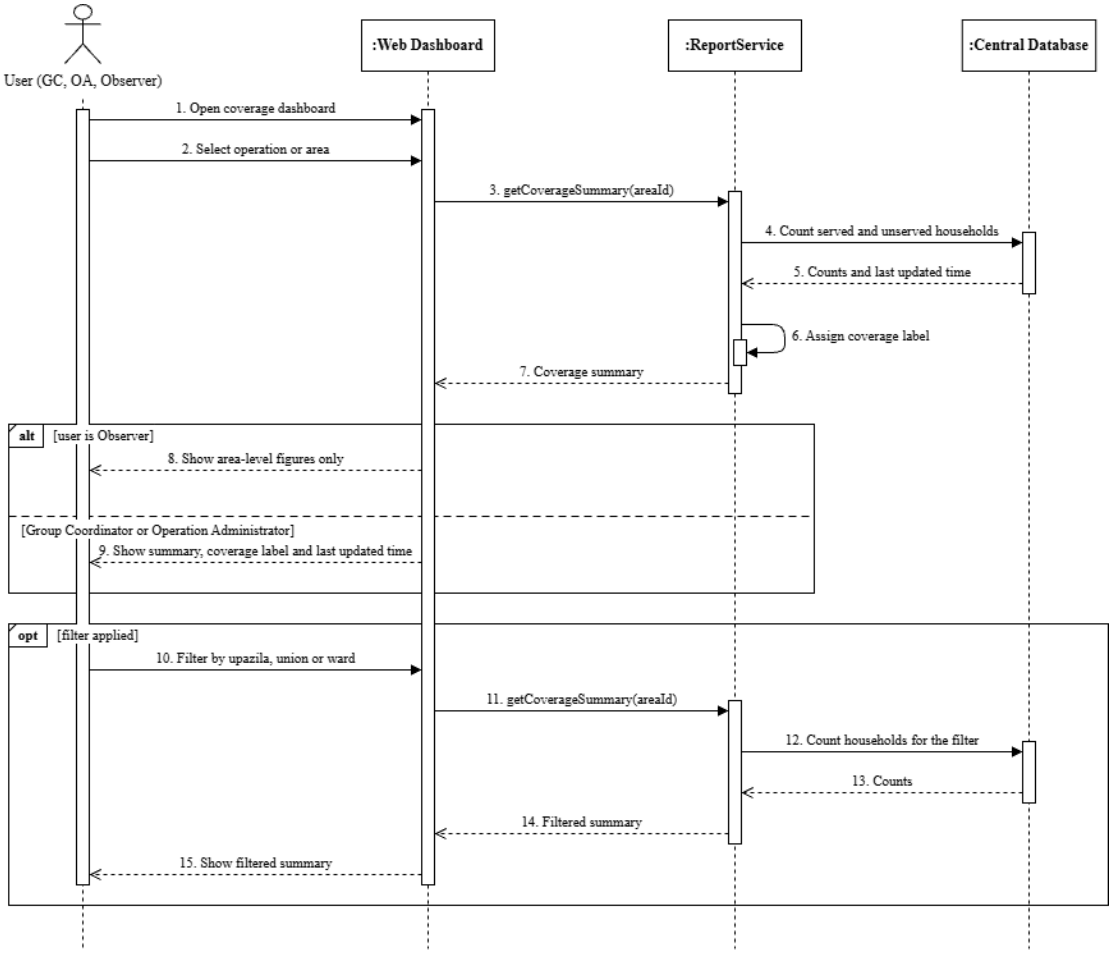


Figure 5.8: Sequence Diagram for View Area Coverage

5.3 State Diagram

5.3.1 State Diagram-01: Pending Record Synchronisation

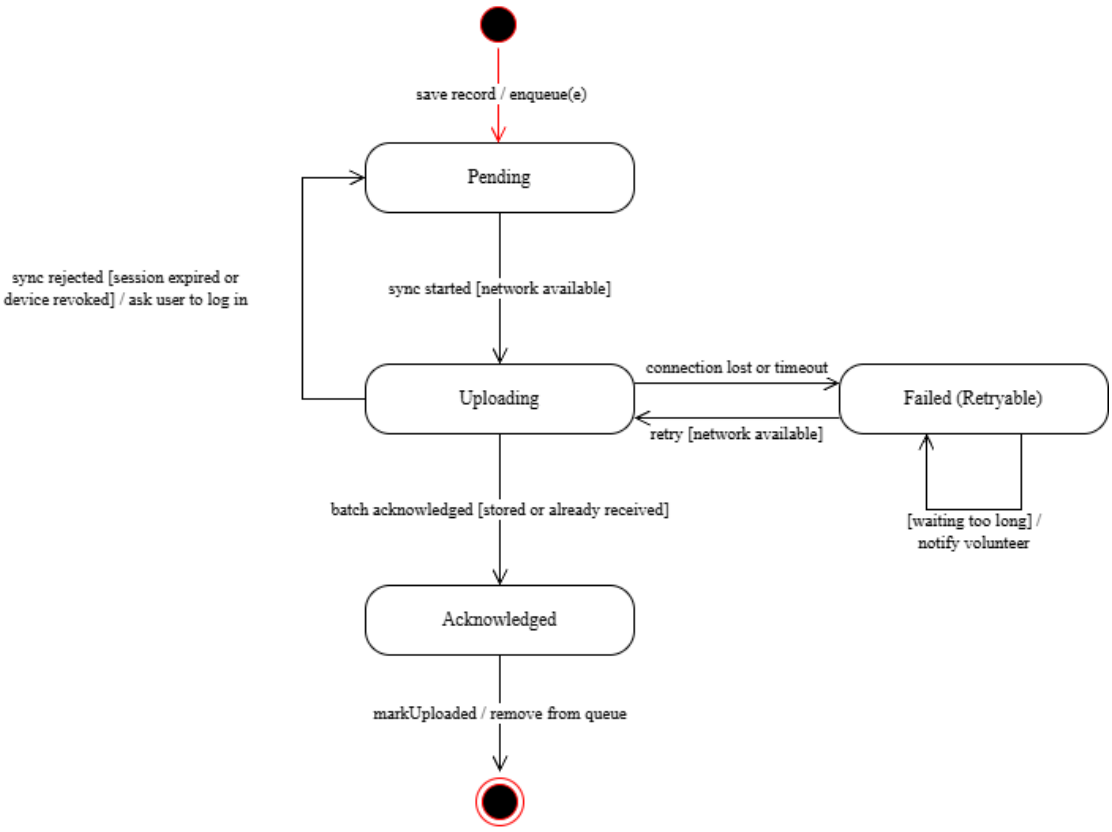


Figure 5.9: State Diagram for Pending Record Synchronisation

5.3.2 State Diagram-02: User Account Status

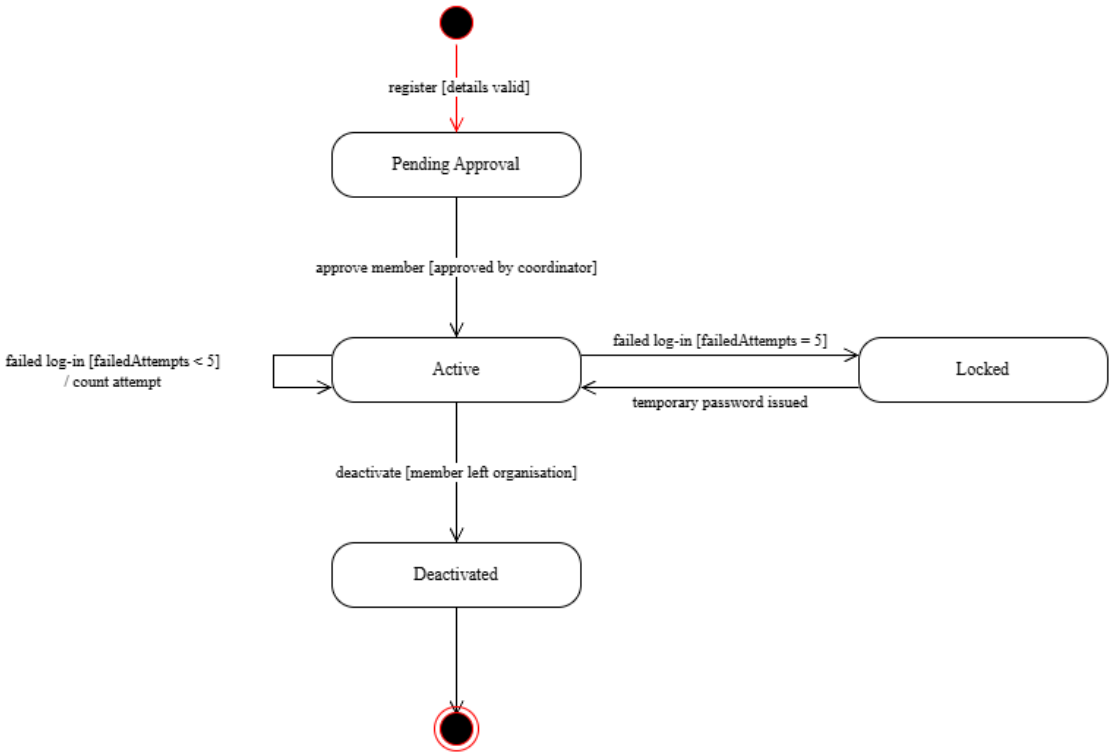


Figure 5.10: State Diagram for User Account Status

5.3.3 State Diagram-03: Organisation Approval Status

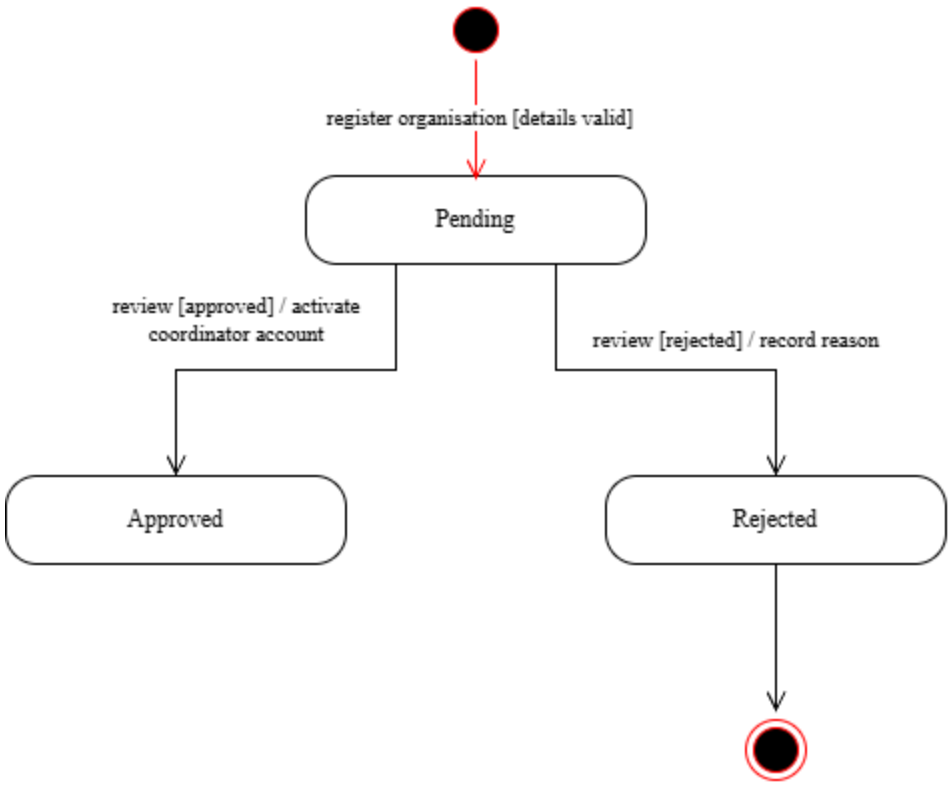


Figure 5.11: State Diagram for Organisation Approval Status

5.3.4 State Diagram-04: Disaster Operation Lifecycle

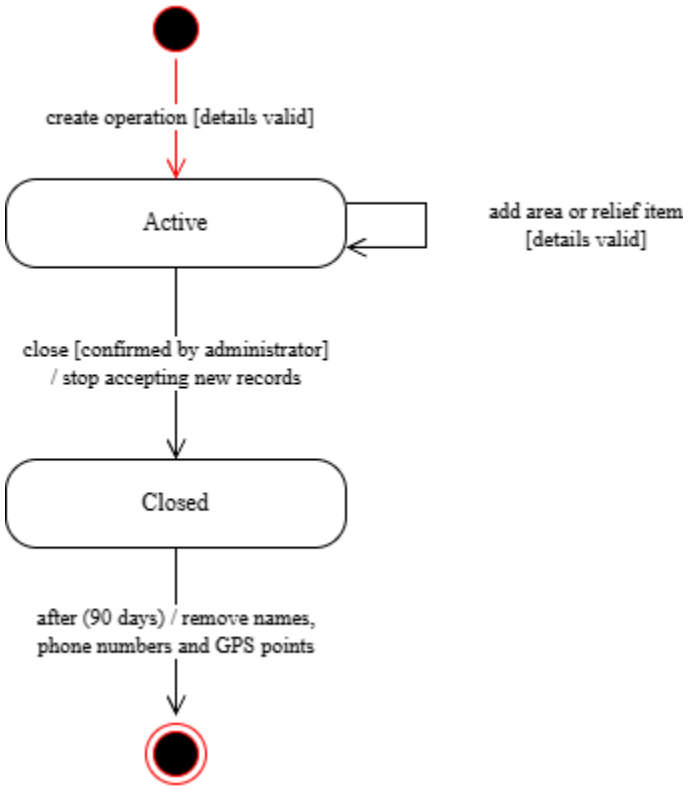


Figure 5.12: State Diagram for Disaster Operation Lifecycle

5.3.5 State Diagram-05: Conflict Review Status

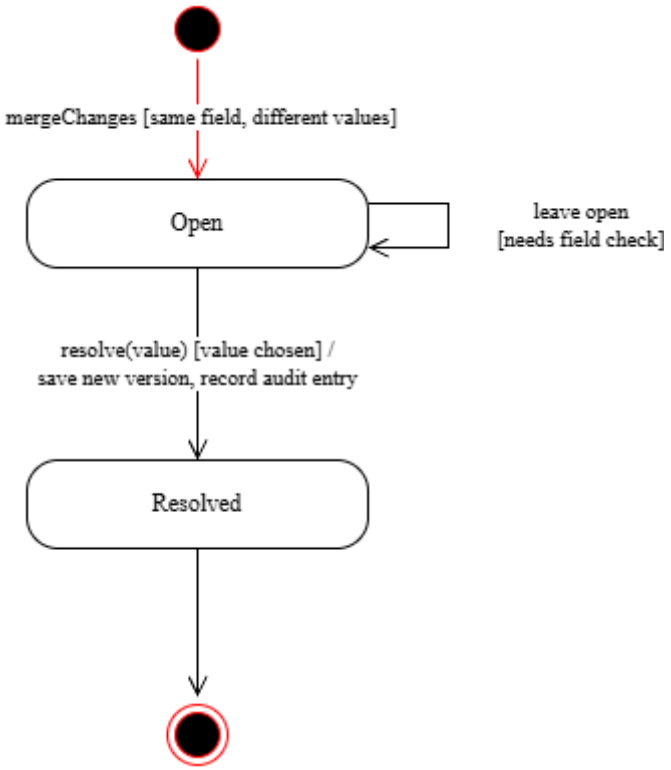


Figure 5.13: State Diagram for Conflict Review Status

5.3.6 State Diagram-06: Duplicate Pair Review Status

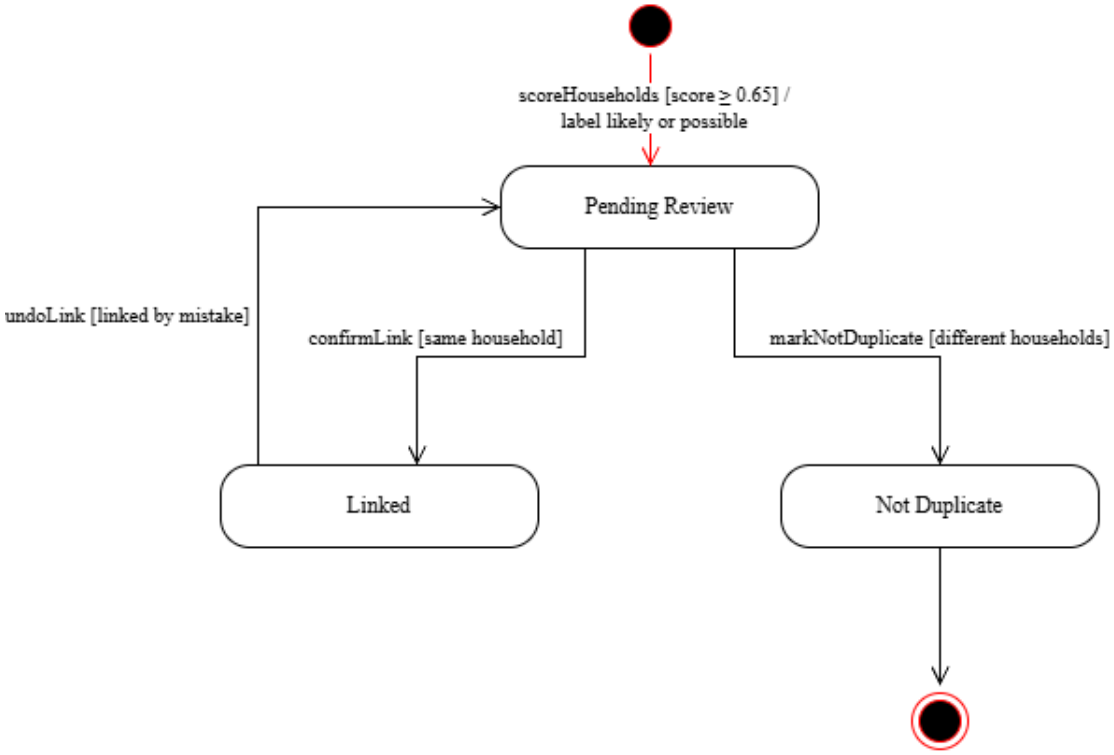


Figure 5.14: State Diagram for Duplicate Pair Review Status

5.4 Entity Relationship Diagram

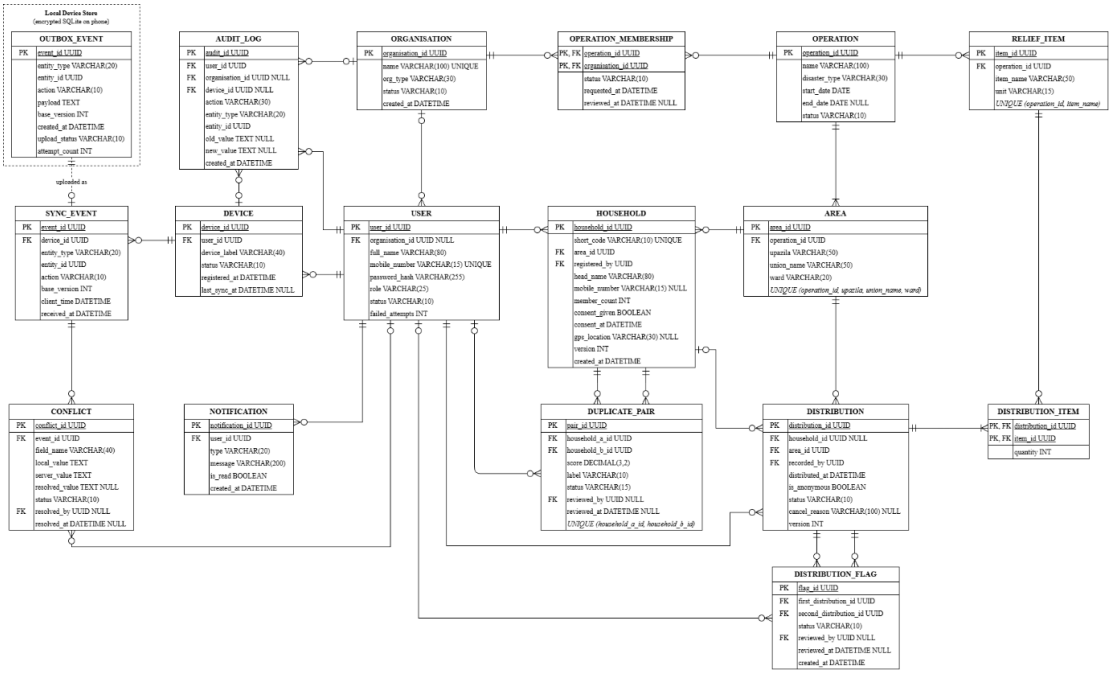


Figure 5.15: Entity Relationship Diagram

5.5 Class Diagram

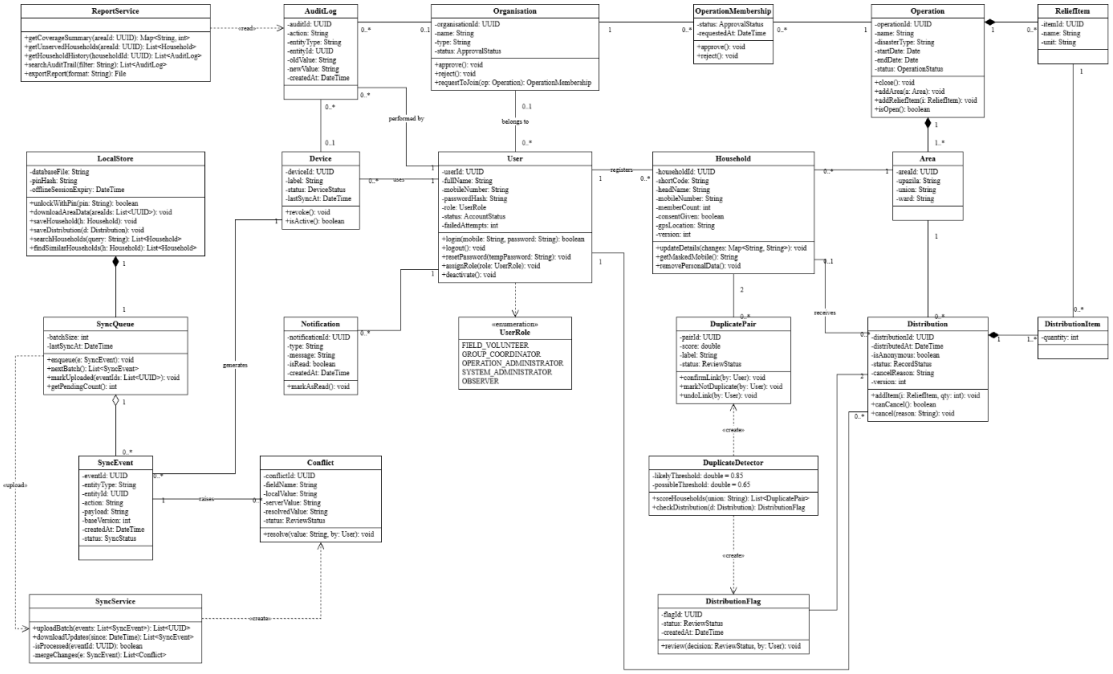


Figure 5.16: Class Diagram